

BFS Traversal

Step 1: Start at E

Legend

Complete

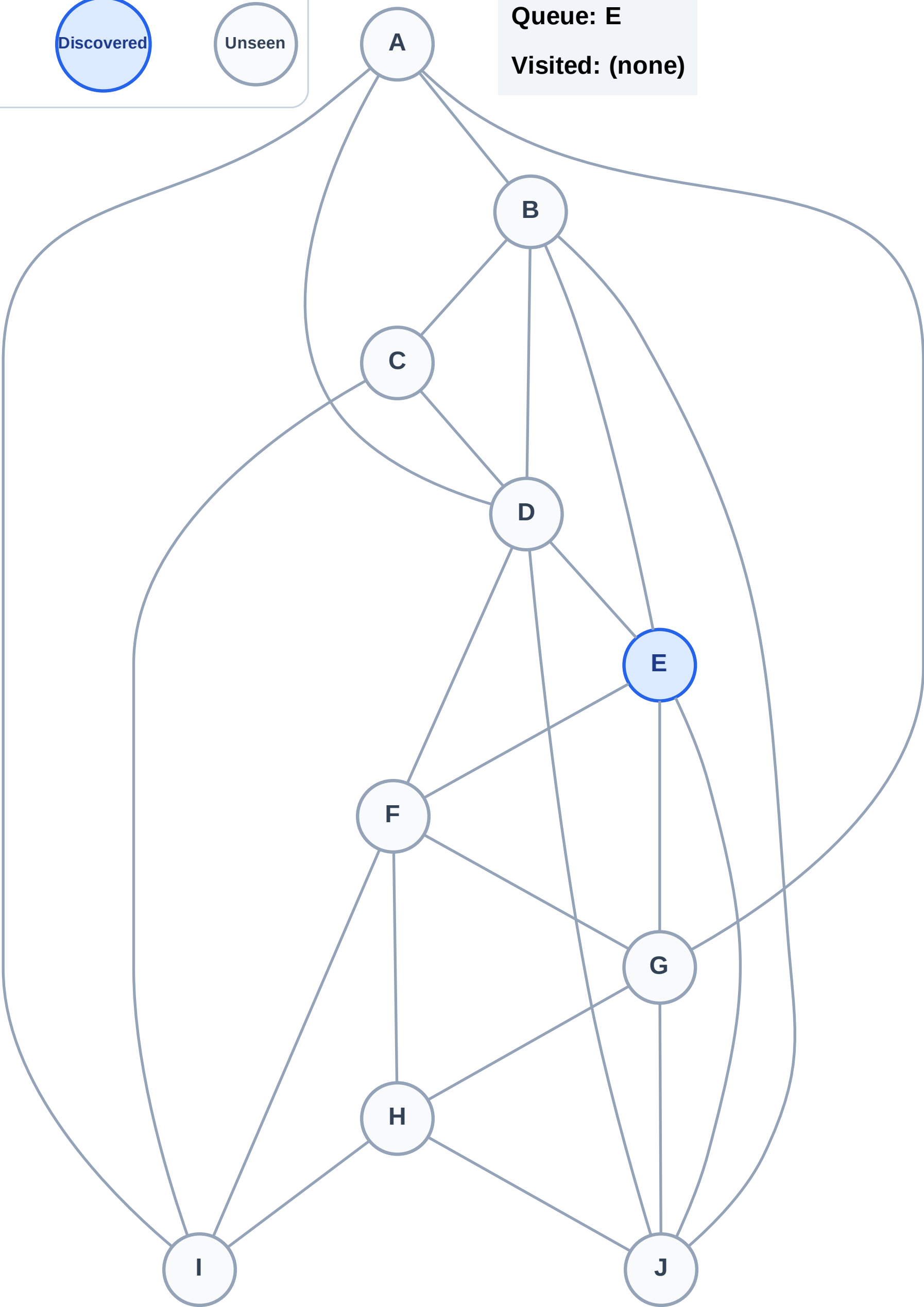
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

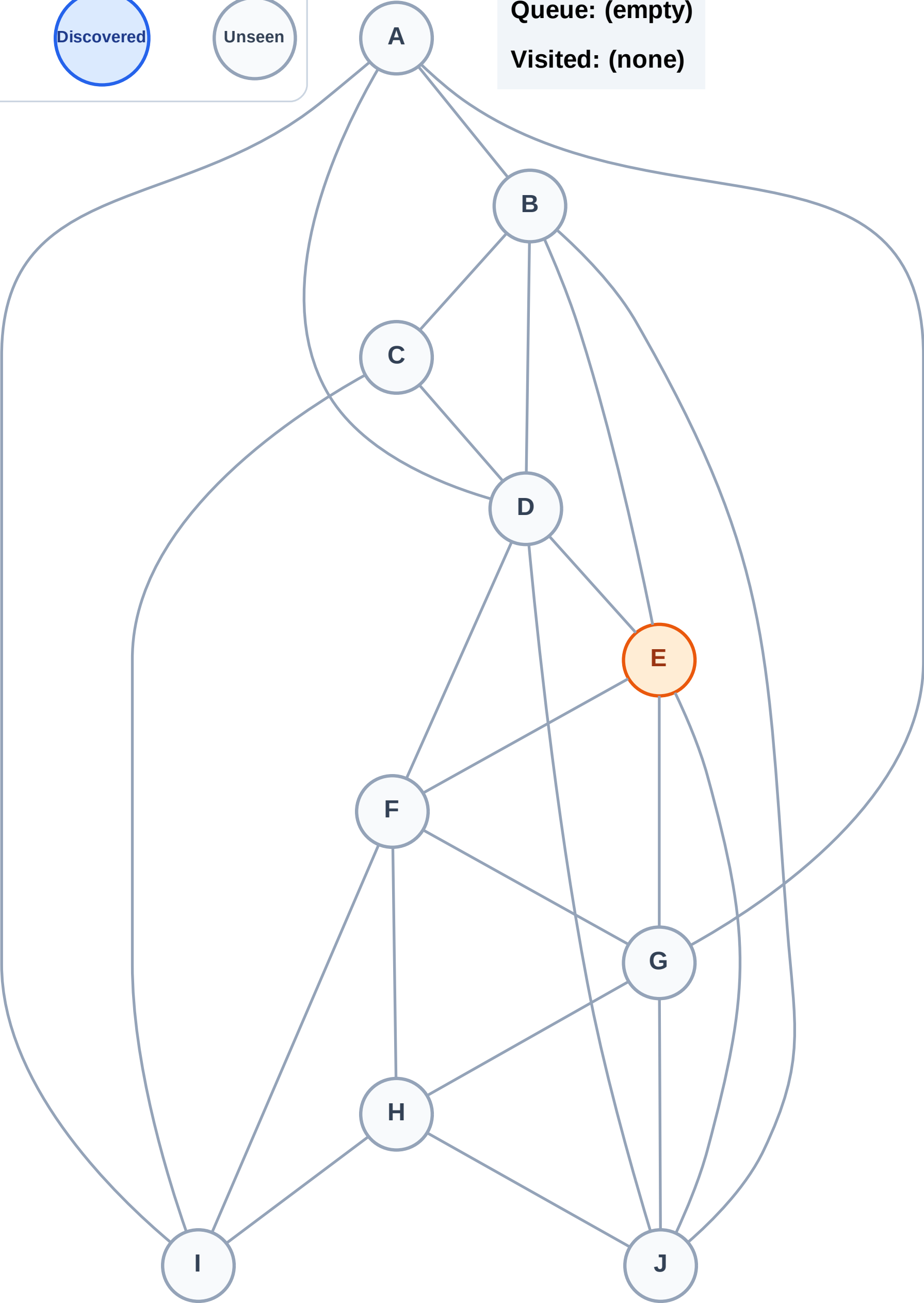
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, D, F, G, J from E

Legend

Complete

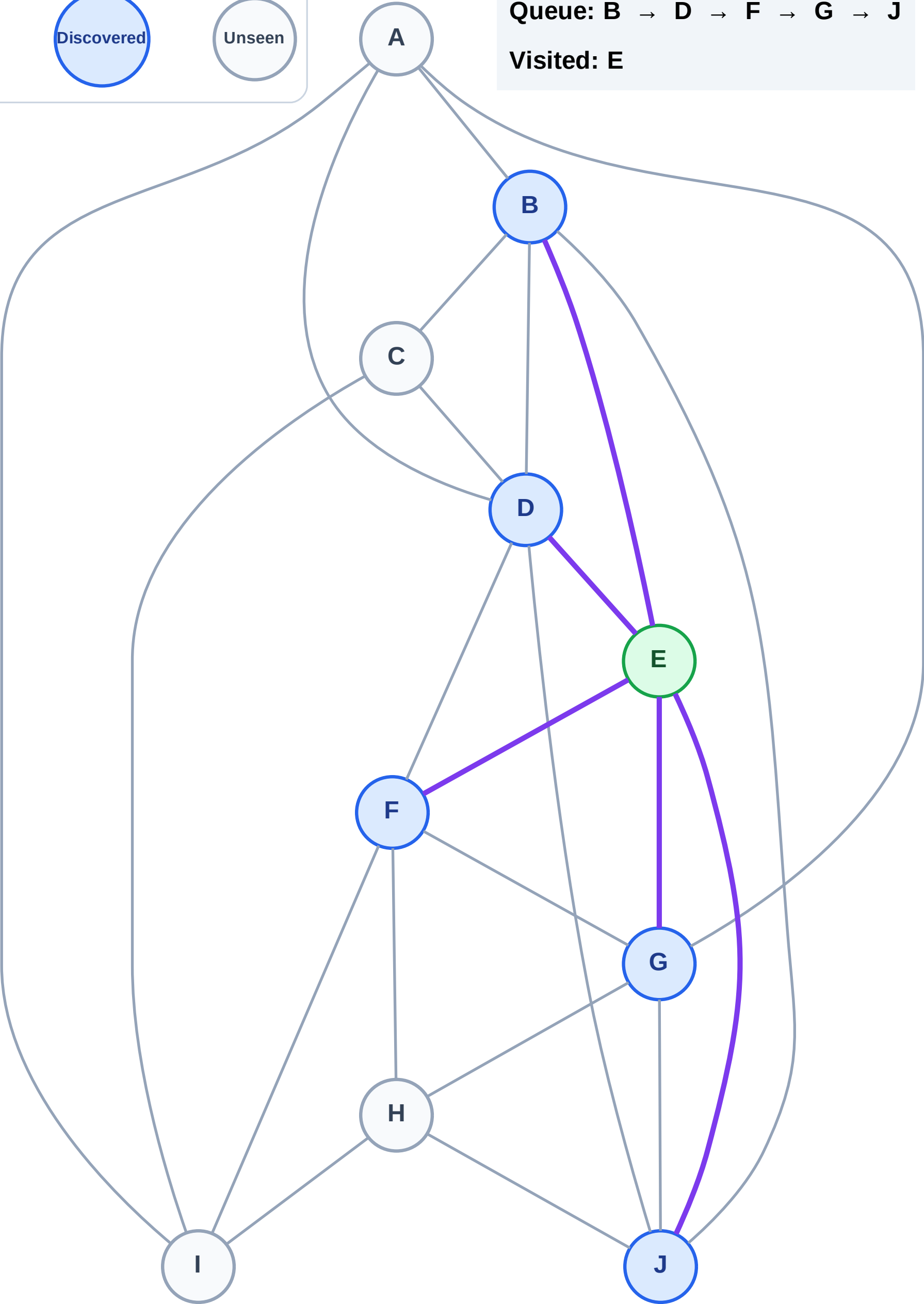
Processing

Discovered

Unseen

Queue: B → D → F → G → J

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

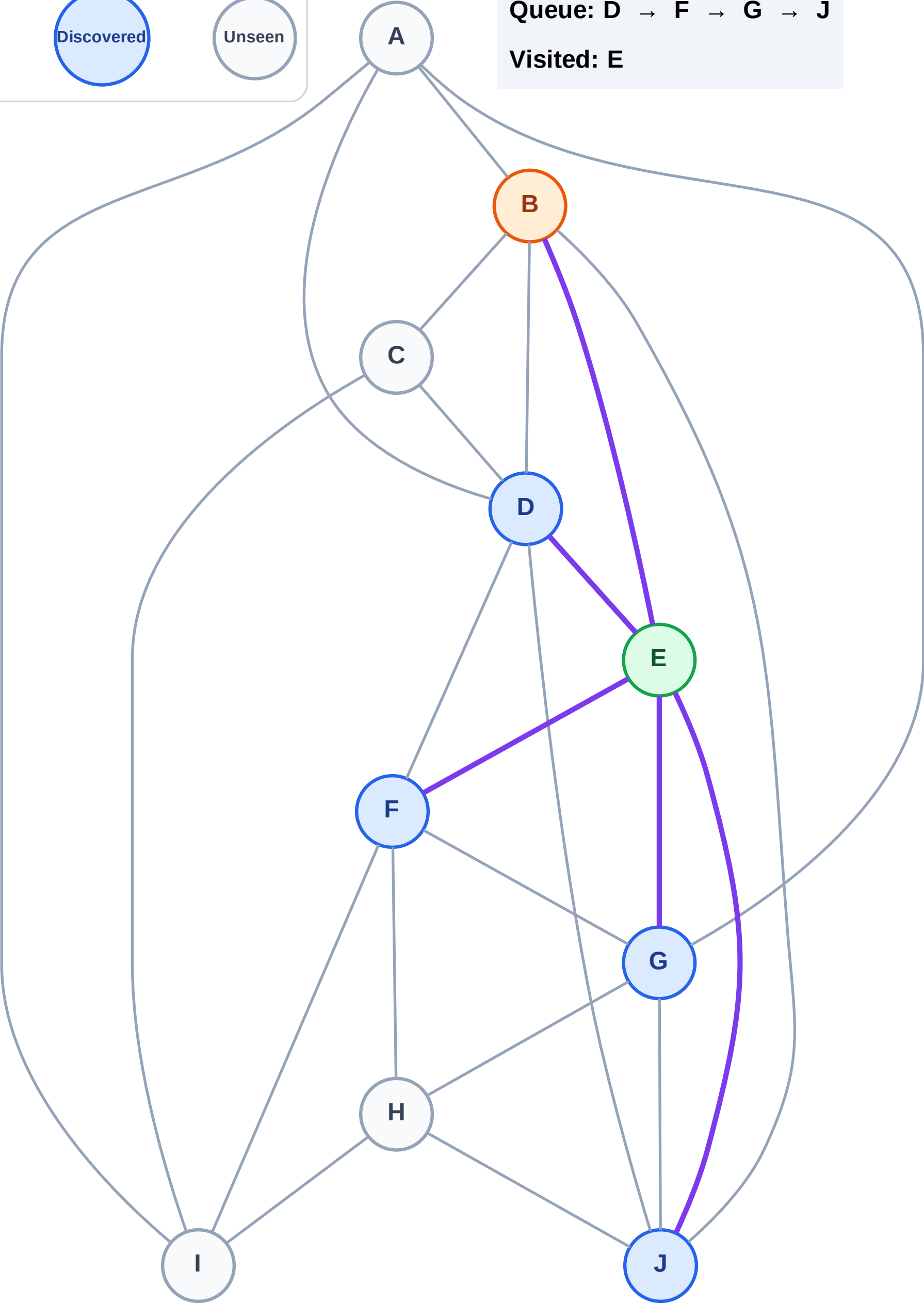
Processing

Discovered

Unseen

Queue: D → F → G → J

Visited: E



BFS Traversal

Step 5: Discover A, C from B

Legend

Complete

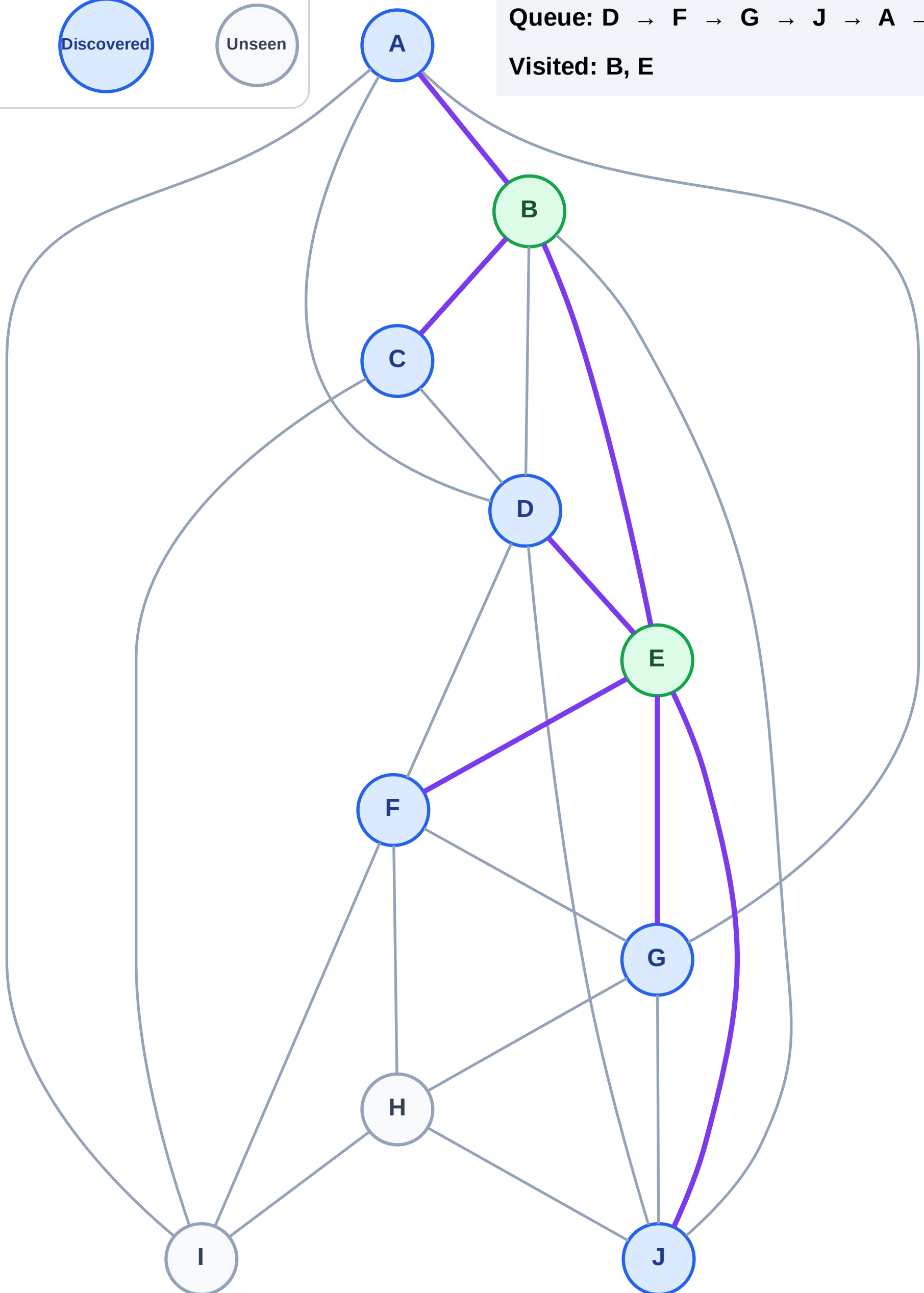
Processing

Discovered

Unseen

Queue: D → F → G → J → A → C

Visited: B, E



BFS Traversal

Step 6: Process node D

Legend

Complete

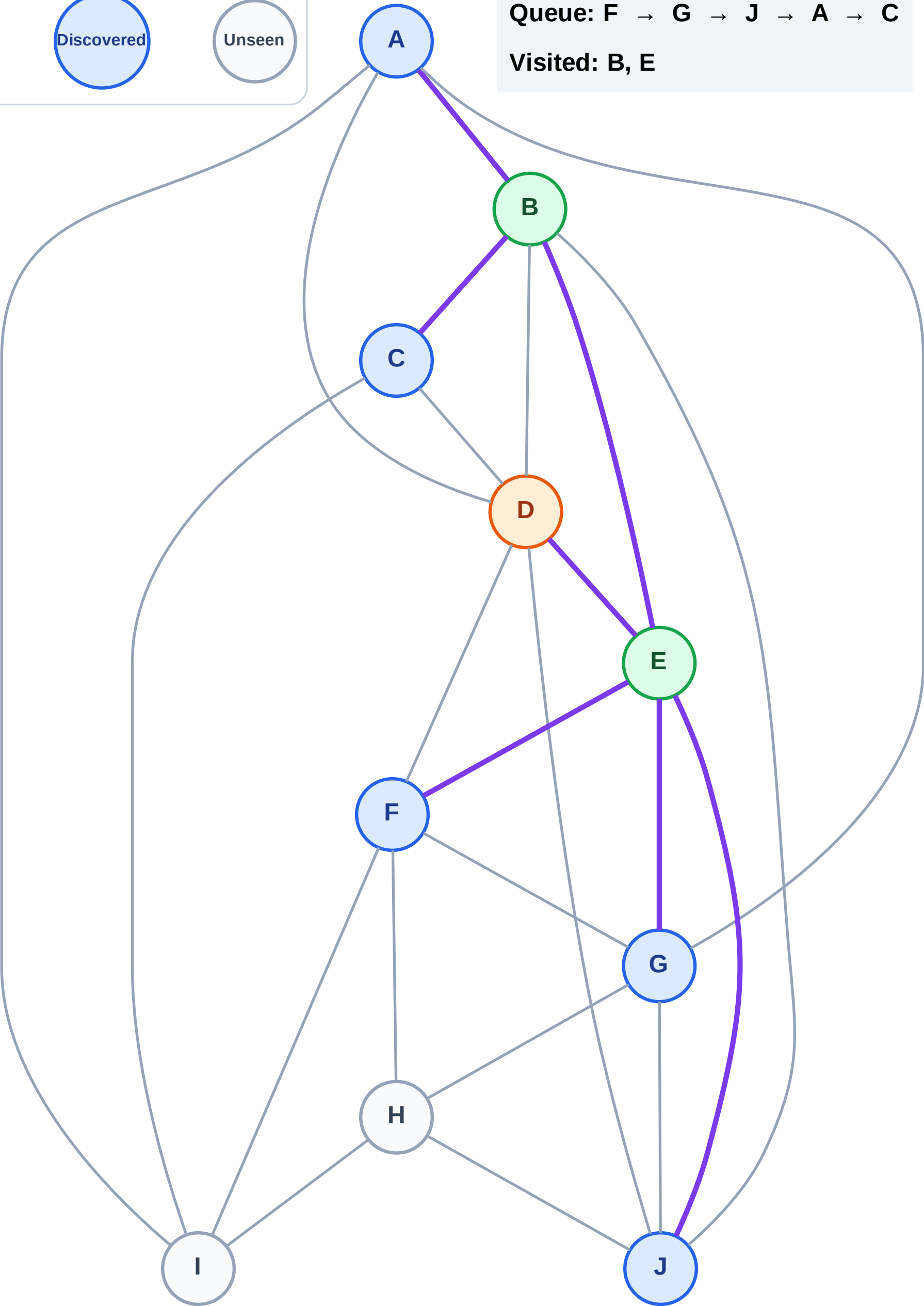
Processing

Discovered

Unseen

Queue: F → G → J → A → C

Visited: B, E



BFS Traversal

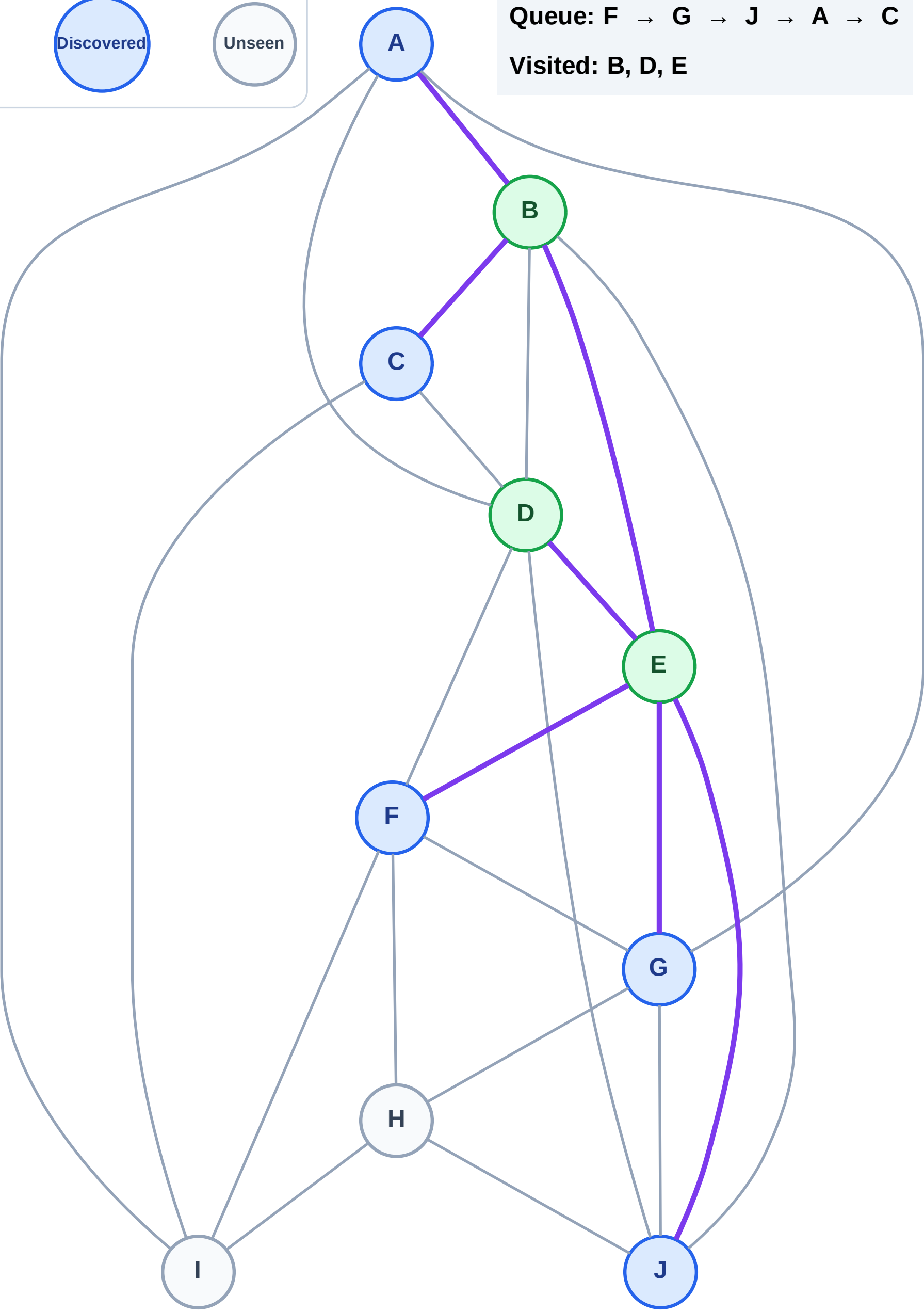
Step 7: No new neighbors from D

Legend



Queue: F → G → J → A → C

Visited: B, D, E



BFS Traversal

Step 8: Process node F

Legend

Complete

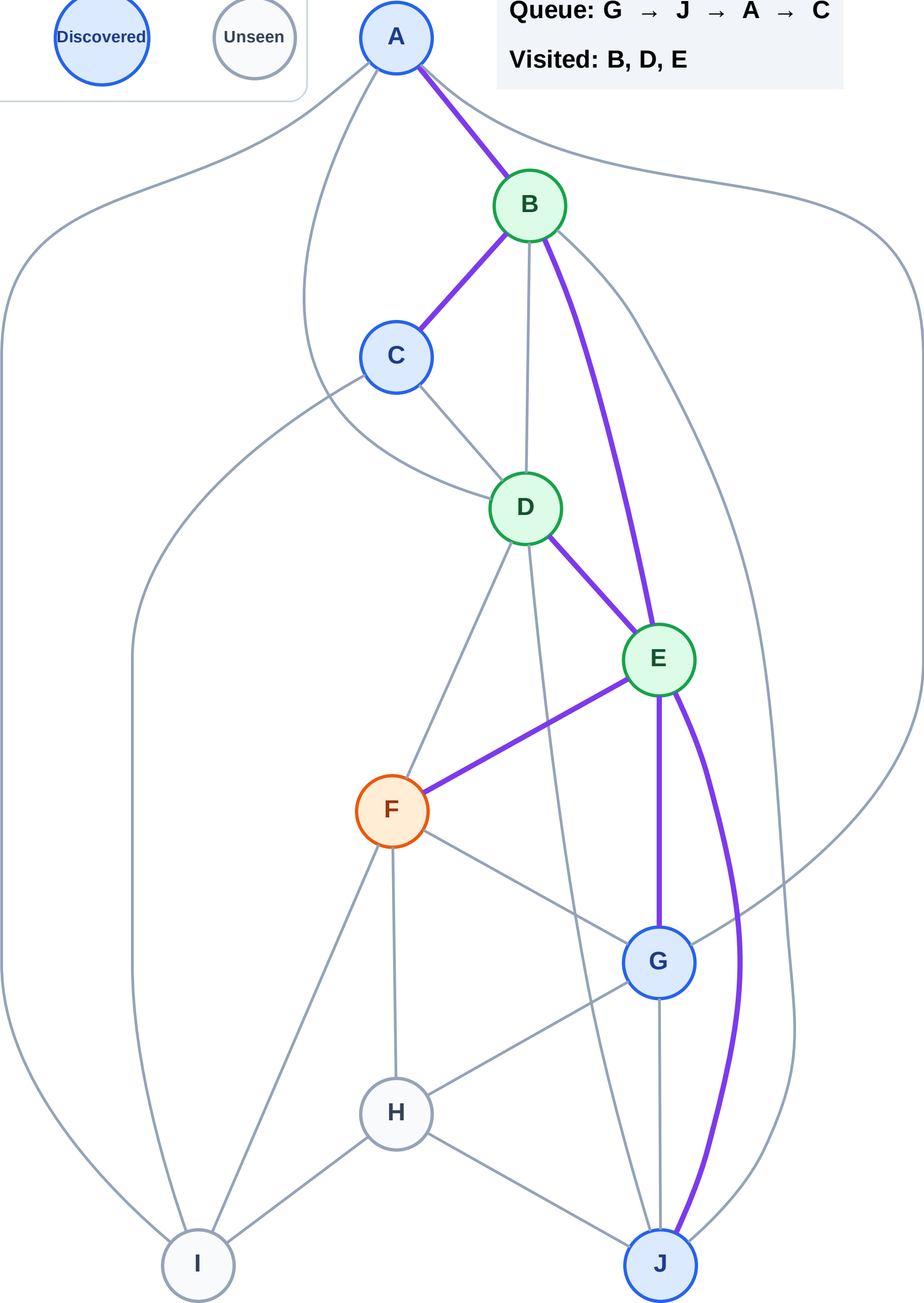
Processing

Discovered

Unseen

Queue: G → J → A → C

Visited: B, D, E



BFS Traversal

Step 9: Discover H, I from F

Legend

Complete

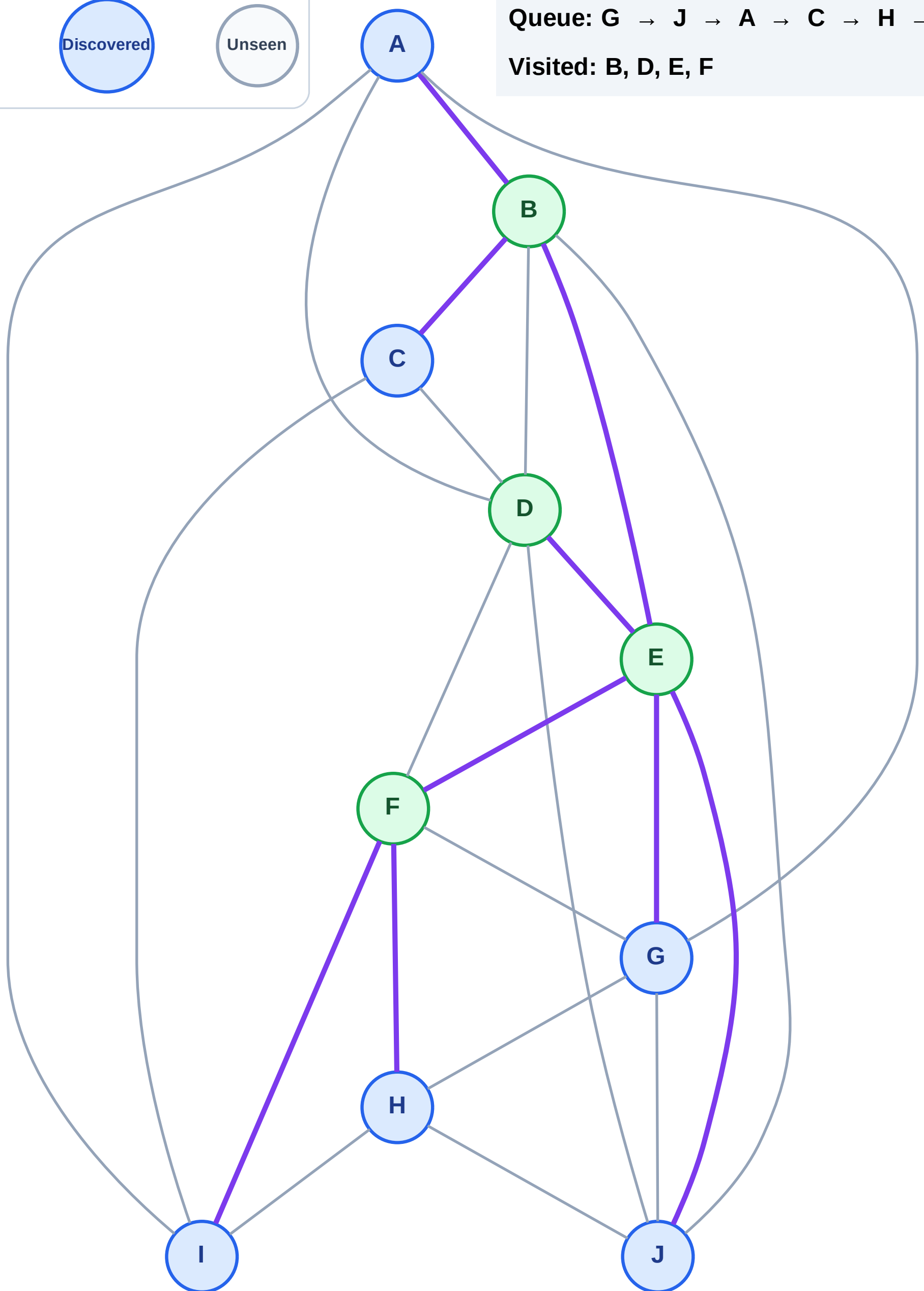
Processing

Discovered

Unseen

Queue: G → J → A → C → H → I

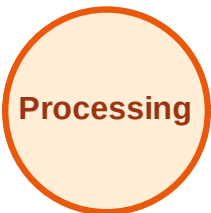
Visited: B, D, E, F



BFS Traversal

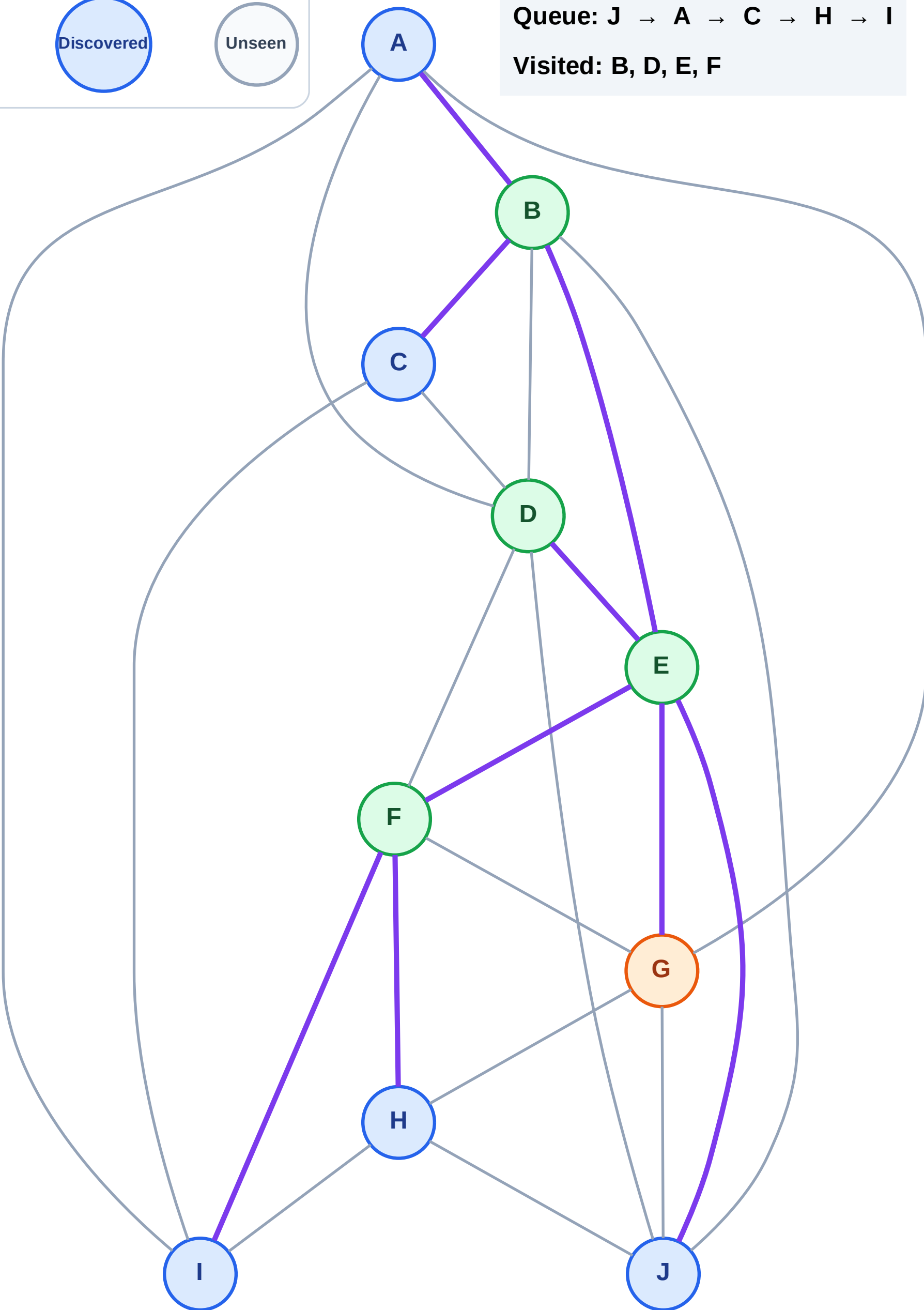
Step 10: Process node G

Legend



Queue: J → A → C → H → I

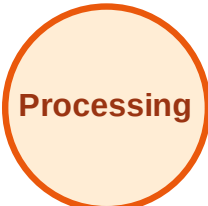
Visited: B, D, E, F



BFS Traversal

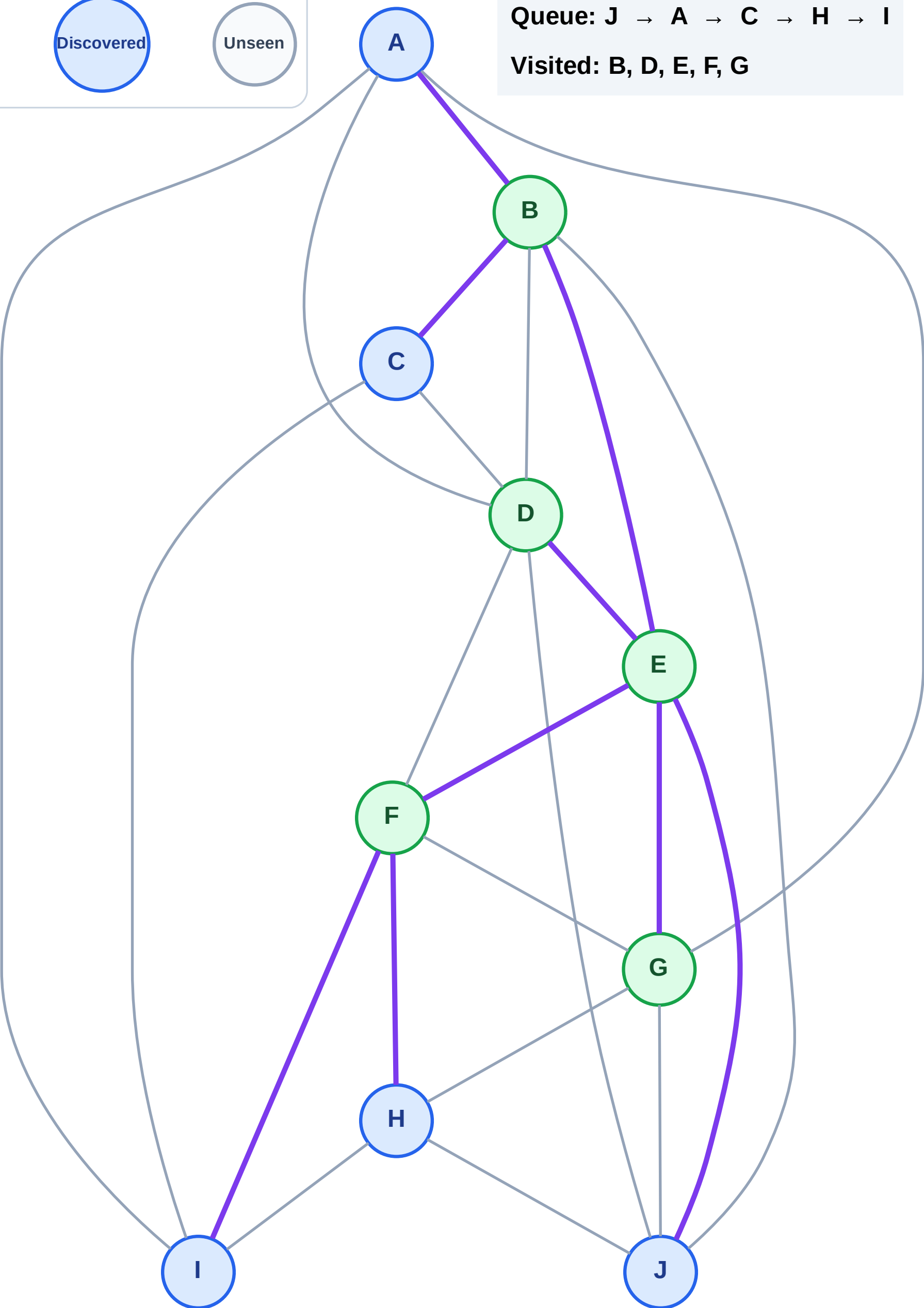
Step 11: No new neighbors from G

Legend



Queue: J → A → C → H → I

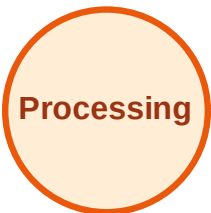
Visited: B, D, E, F, G



BFS Traversal

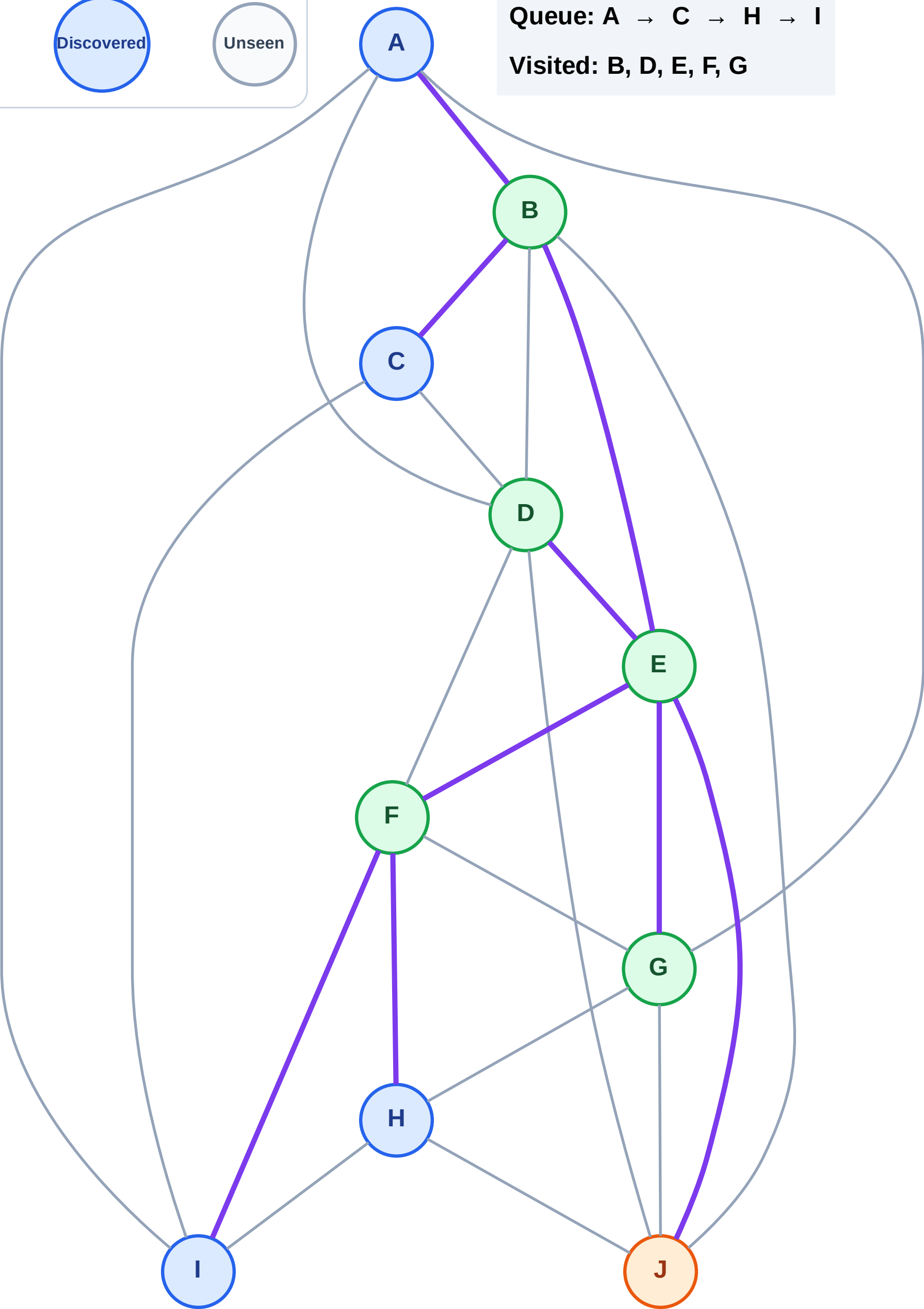
Step 12: Process node J

Legend



Queue: A → C → H → I

Visited: B, D, E, F, G



BFS Traversal

Step 13: No new neighbors from J

Legend

Complete

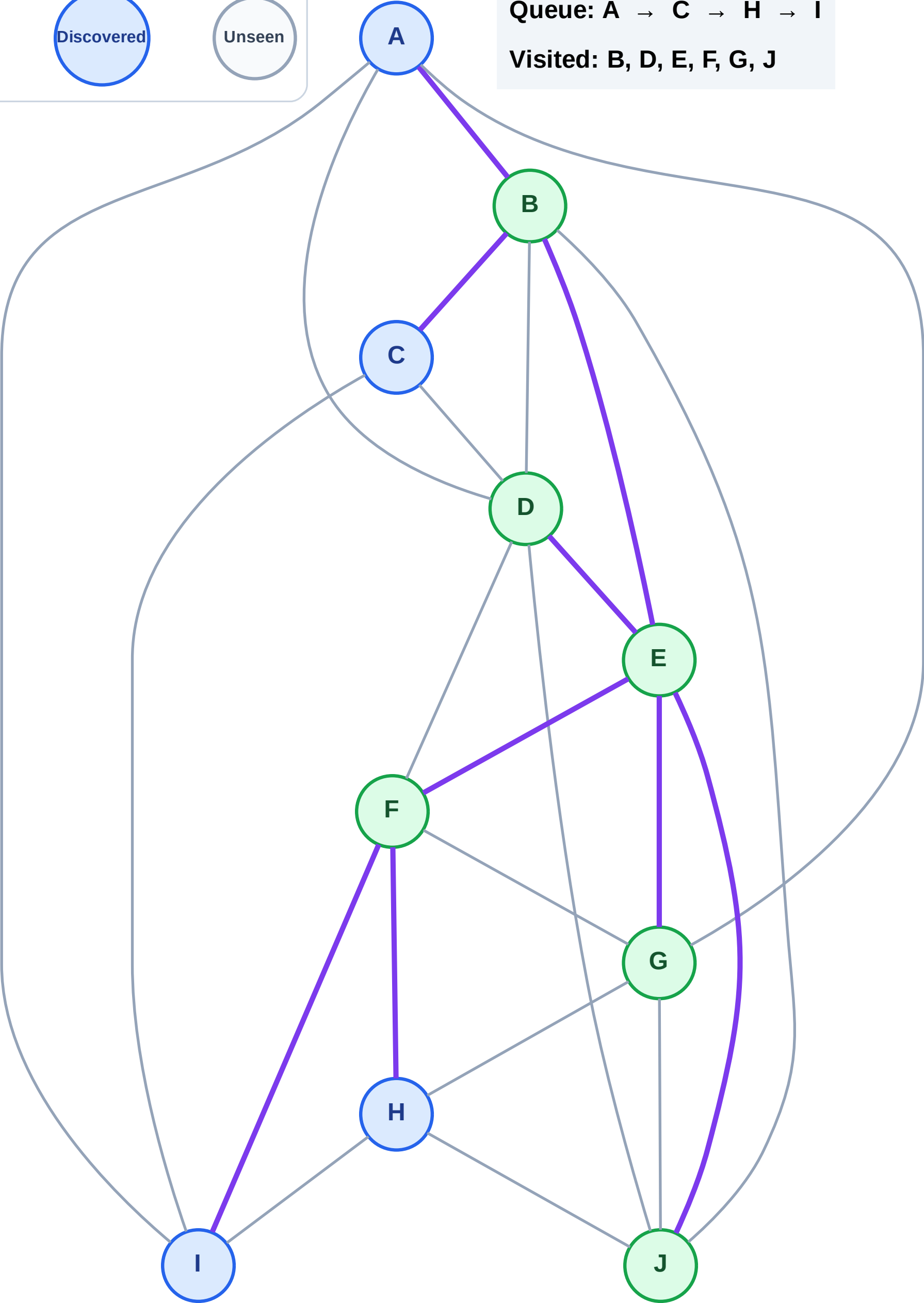
Processing

Discovered

Unseen

Queue: A → C → H → I

Visited: B, D, E, F, G, J



BFS Traversal

Step 14: Process node A

Legend

Complete

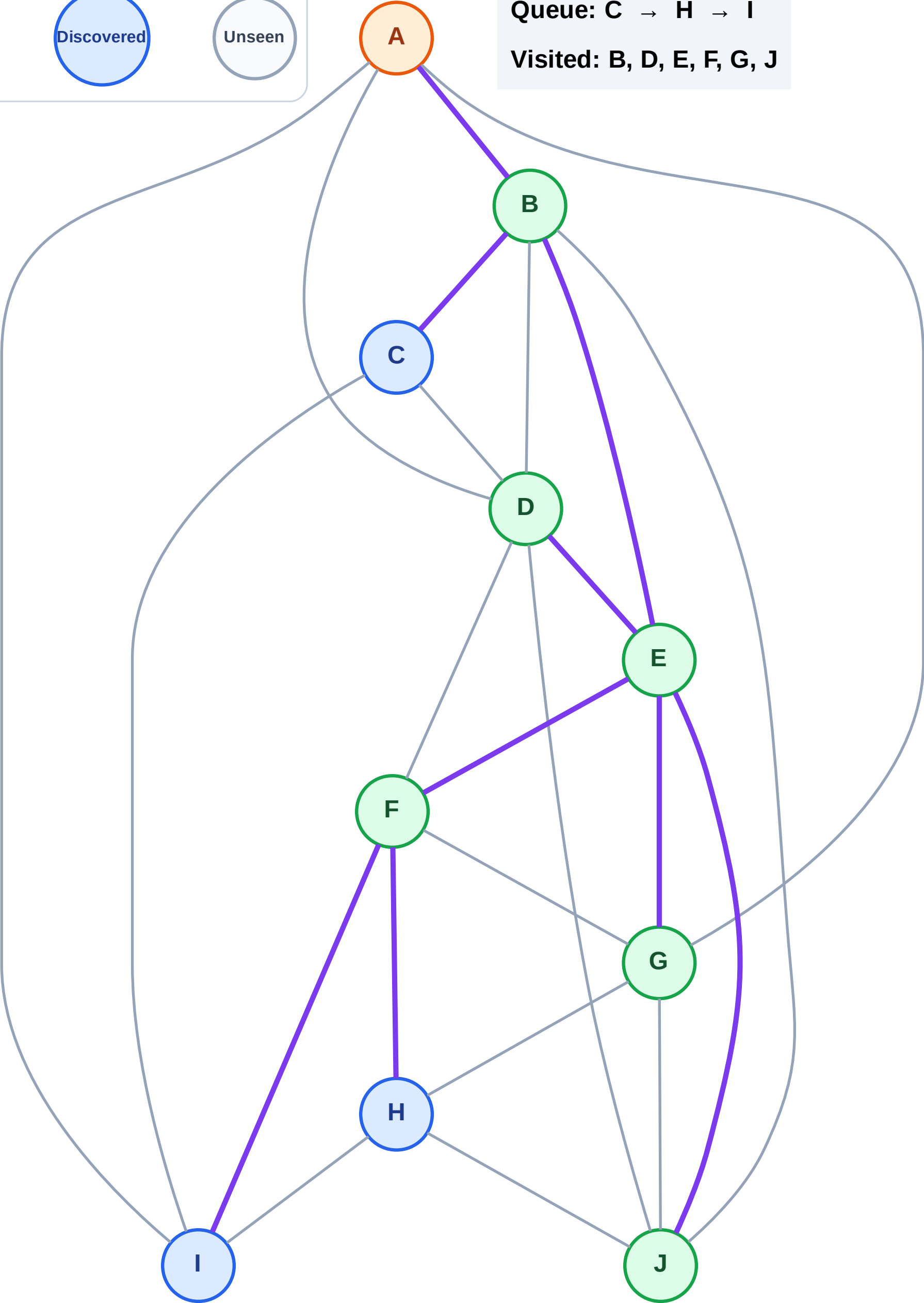
Processing

Discovered

Unseen

Queue: C → H → I

Visited: B, D, E, F, G, J



BFS Traversal

Step 15: No new neighbors from A

Legend

Complete

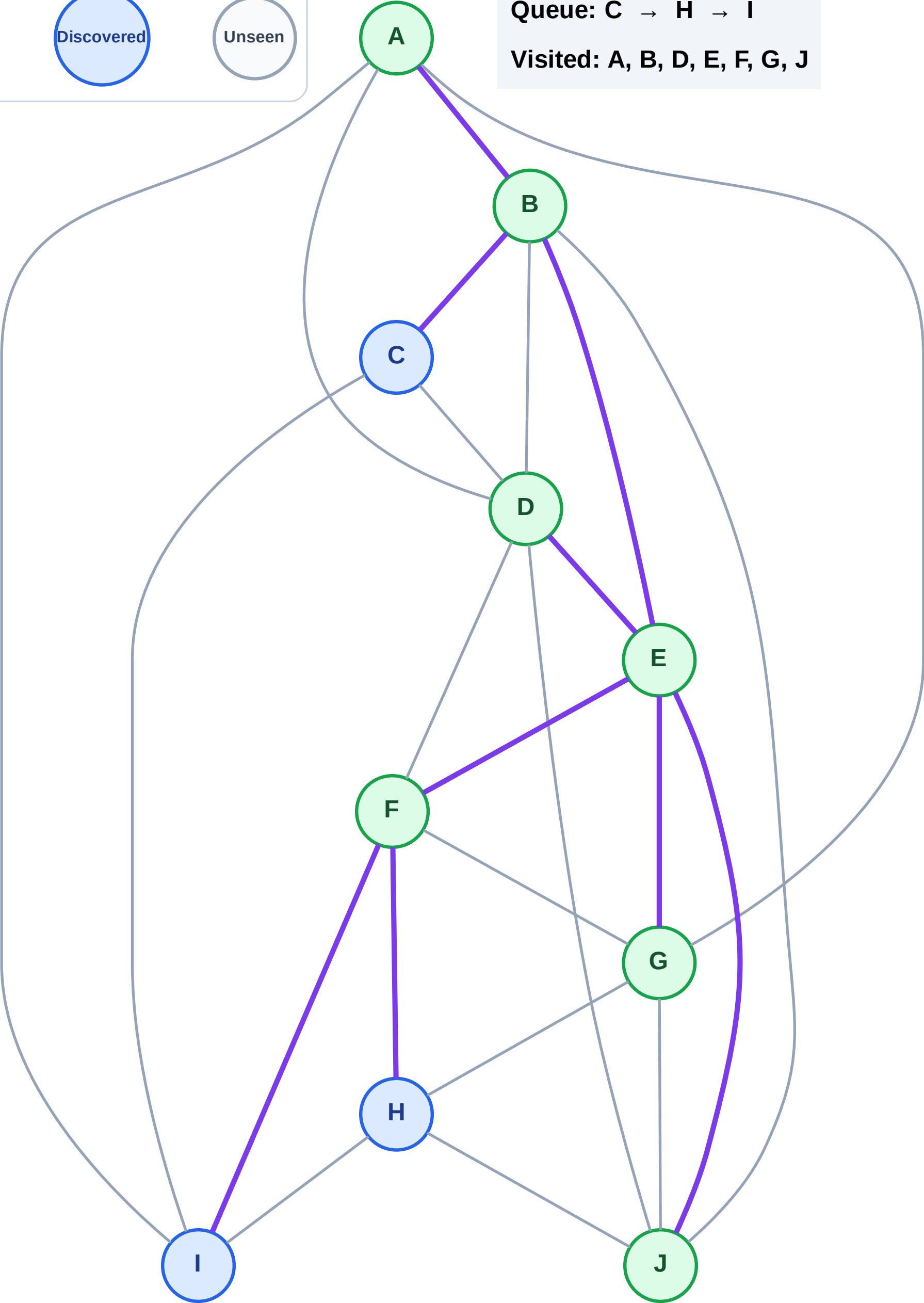
Processing

Discovered

Unseen

Queue: C → H → I

Visited: A, B, D, E, F, G, J



BFS Traversal

Step 16: Process node C

Legend

Complete

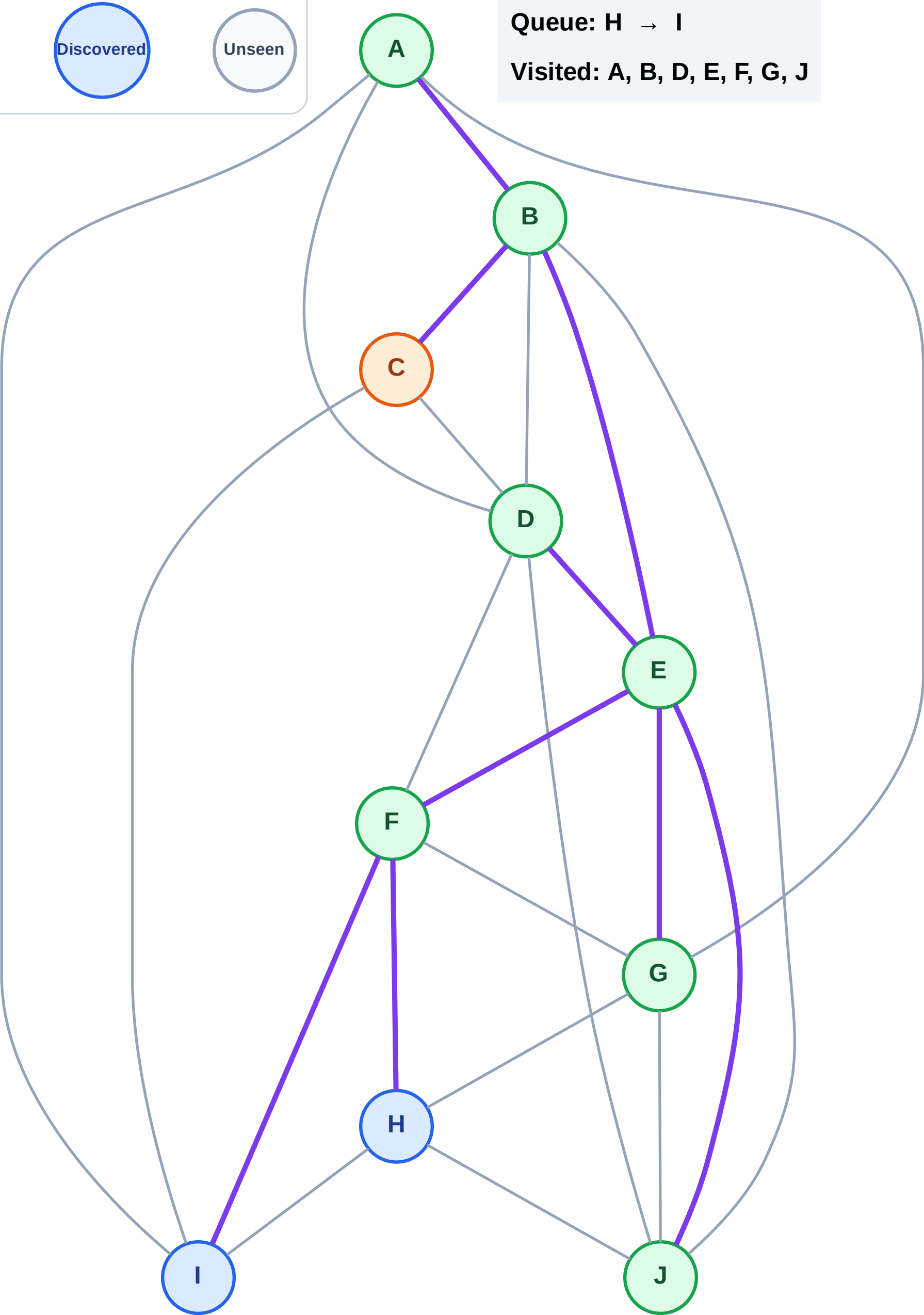
Processing

Discovered

Unseen

Queue: H → I

Visited: A, B, D, E, F, G, J



BFS Traversal

Step 17: No new neighbors from C

Legend

Complete

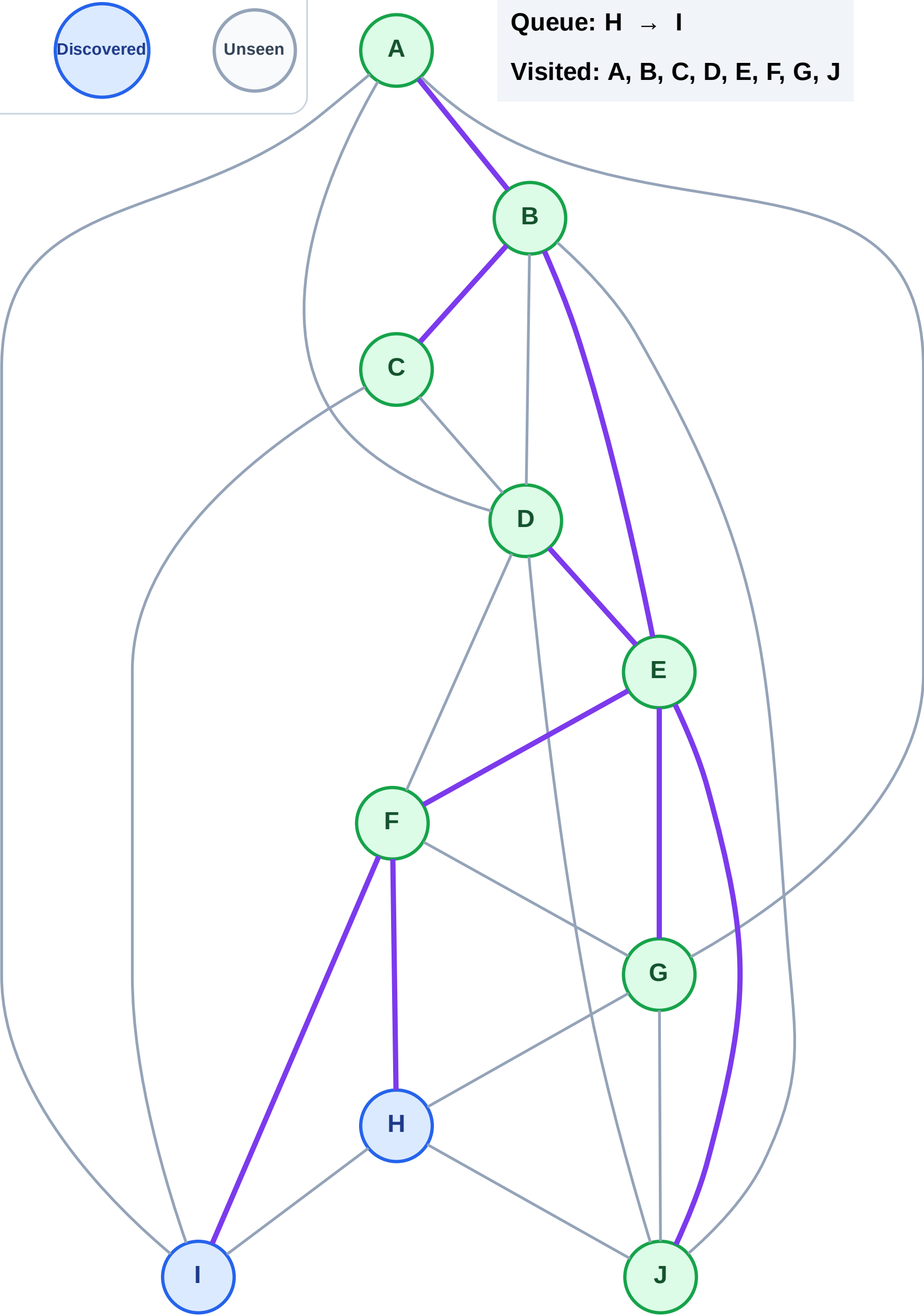
Processing

Discovered

Unseen

Queue: H → I

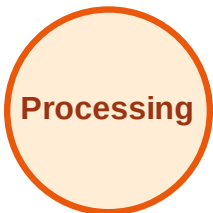
Visited: A, B, C, D, E, F, G, J



BFS Traversal

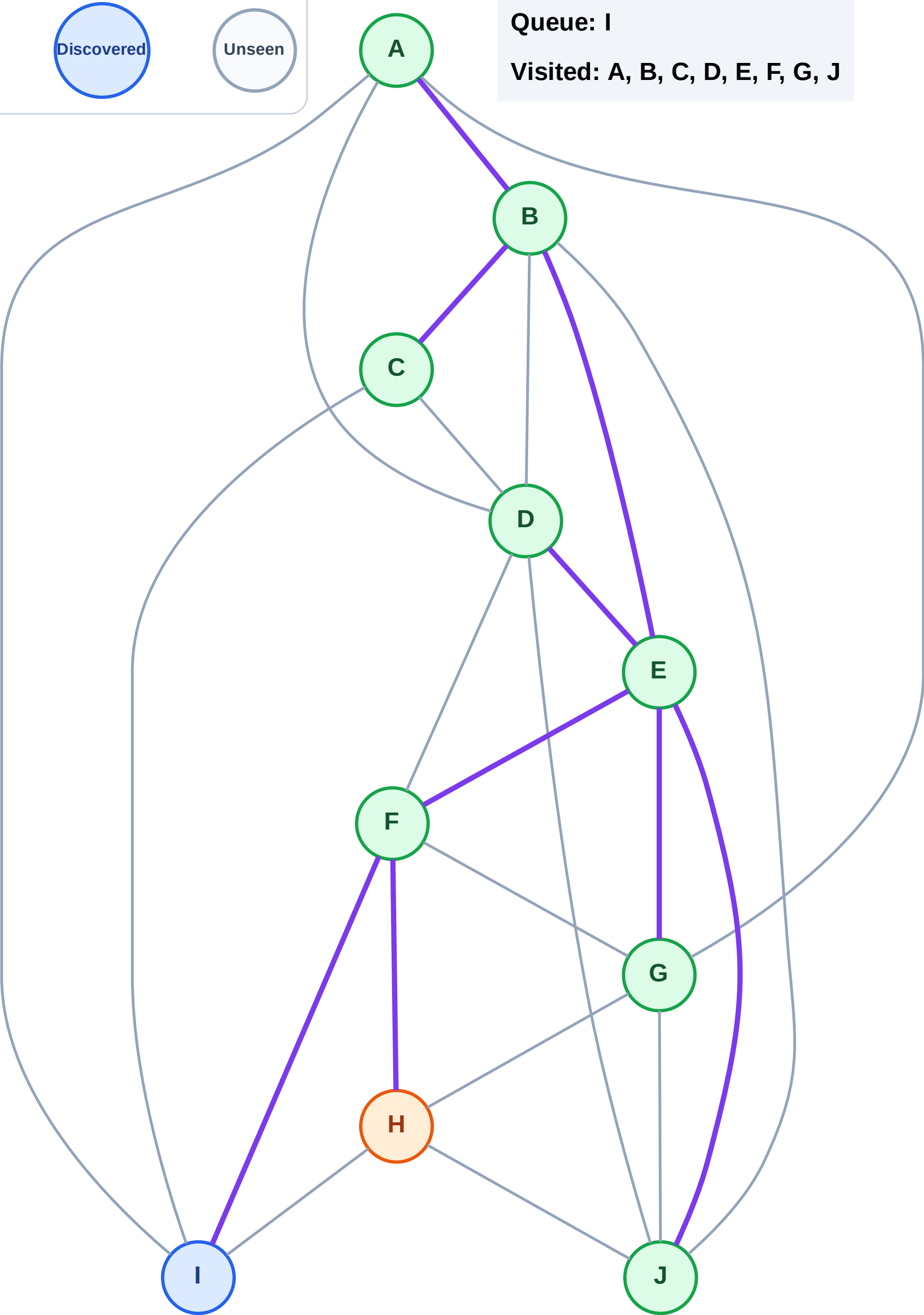
Step 18: Process node H

Legend



Queue: I

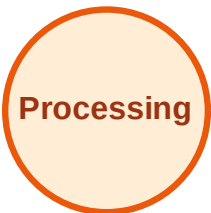
Visited: A, B, C, D, E, F, G, J



BFS Traversal

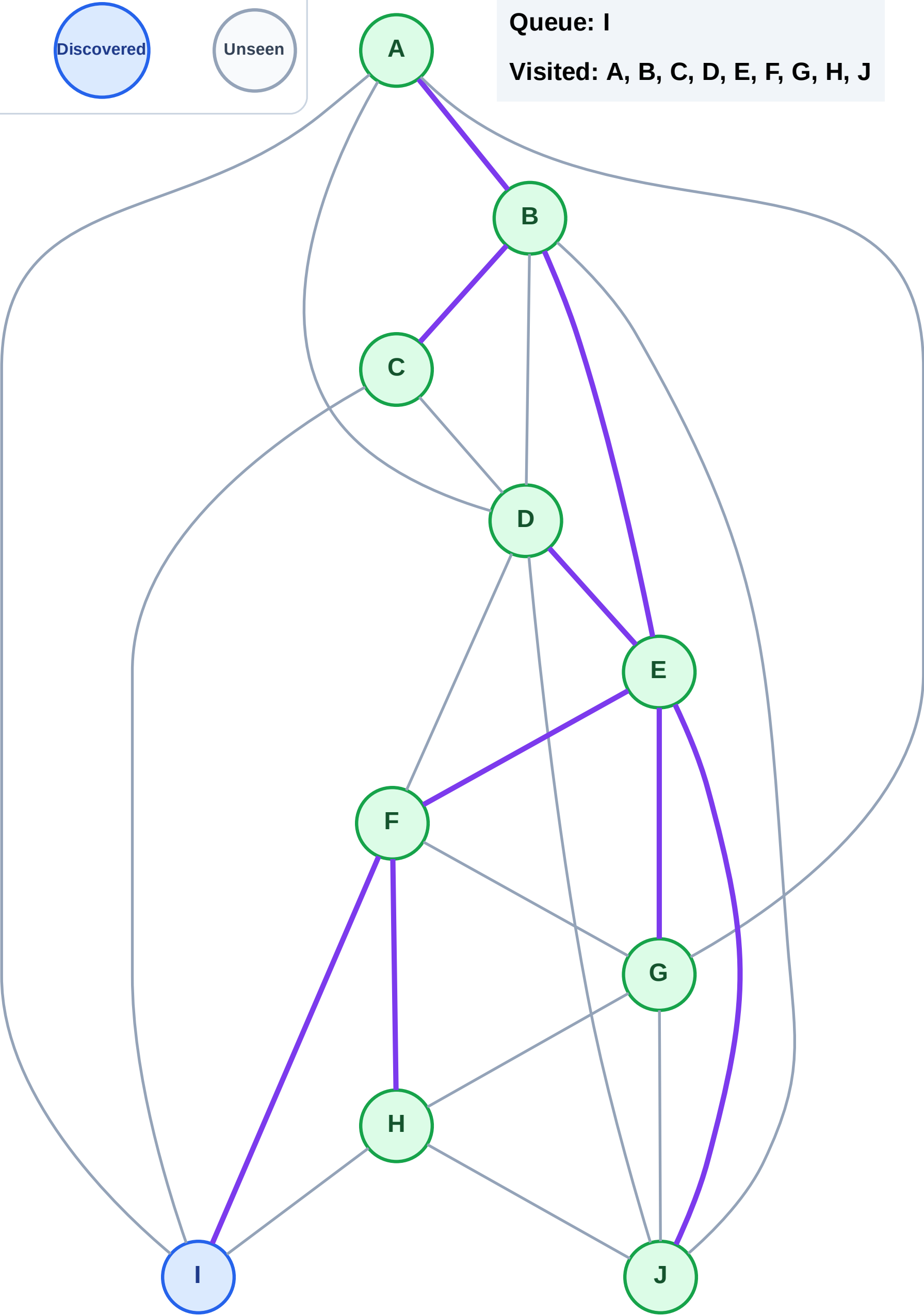
Step 19: No new neighbors from H

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

Step 20: Process node I

Legend

Complete

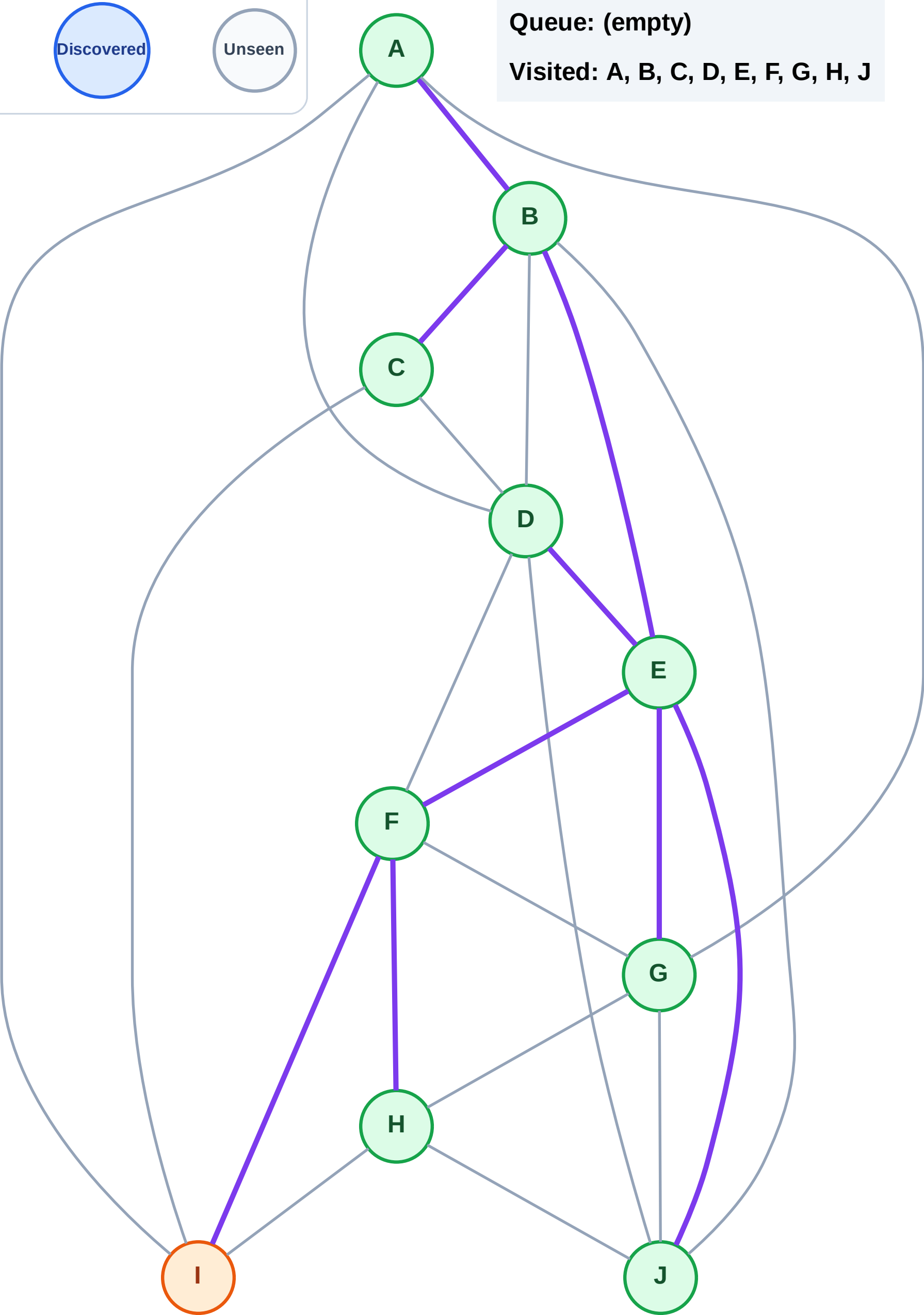
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

Step 21: No new neighbors from I

Legend

Complete

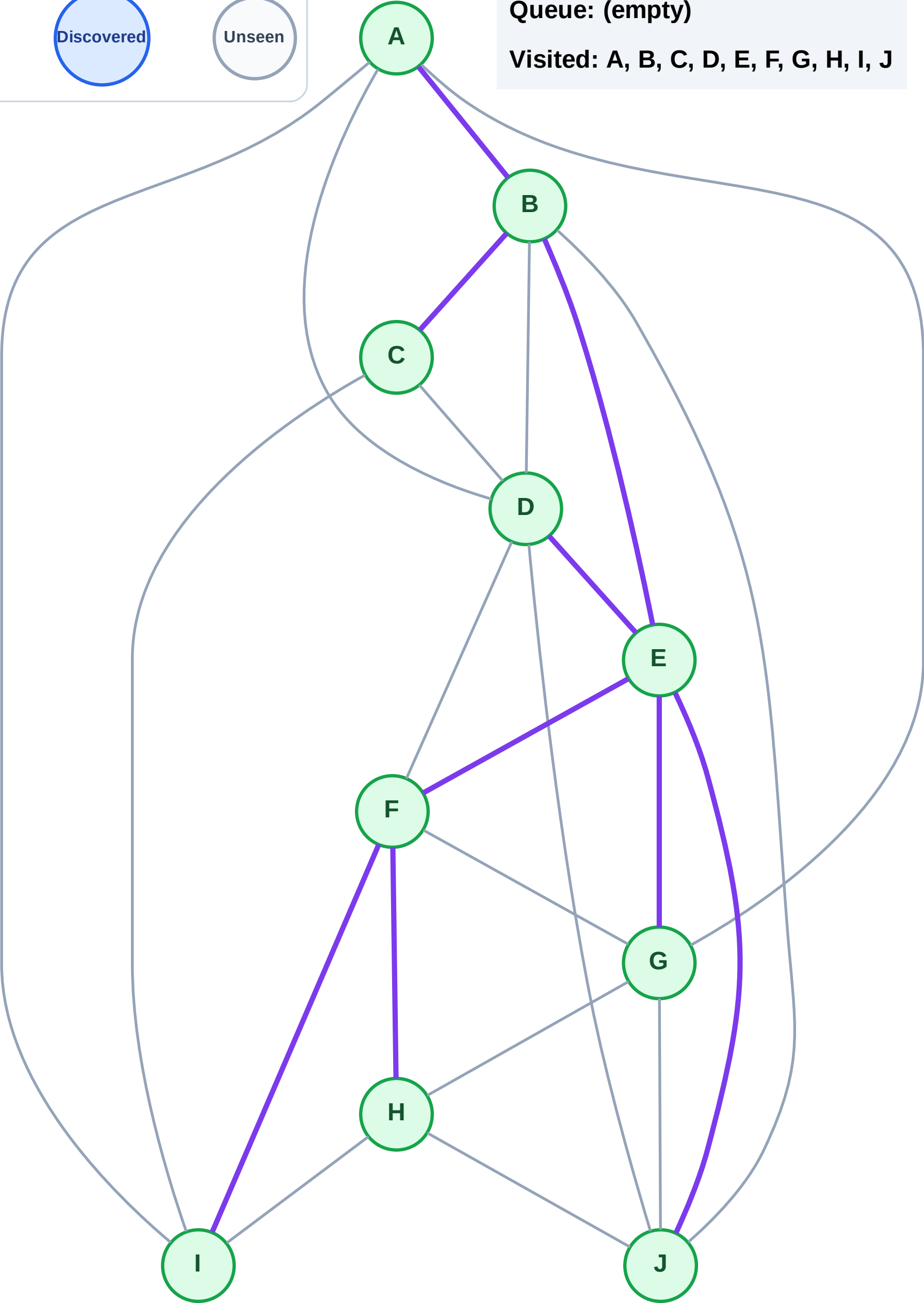
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

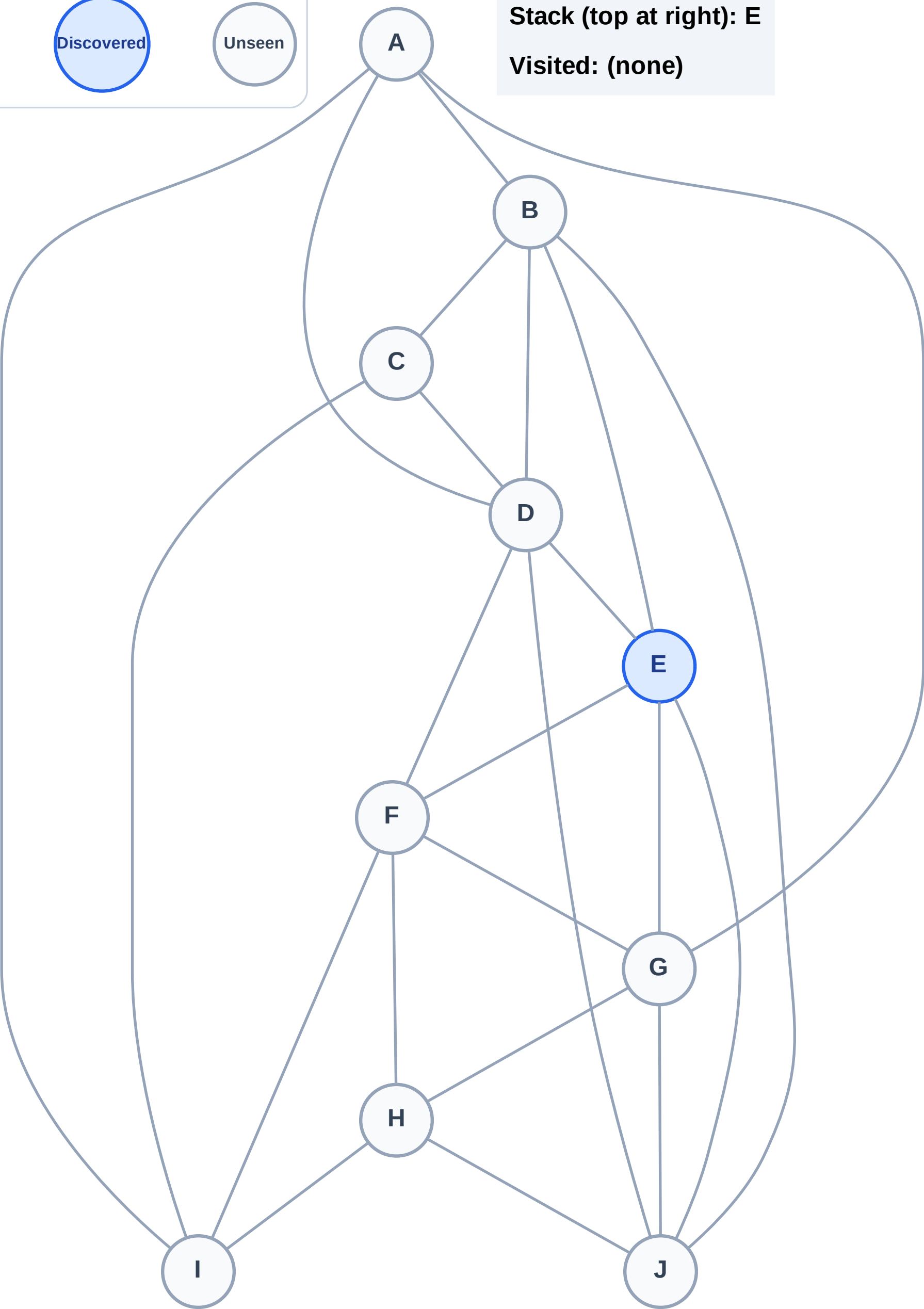
Processing

Discovered

Unseen

Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

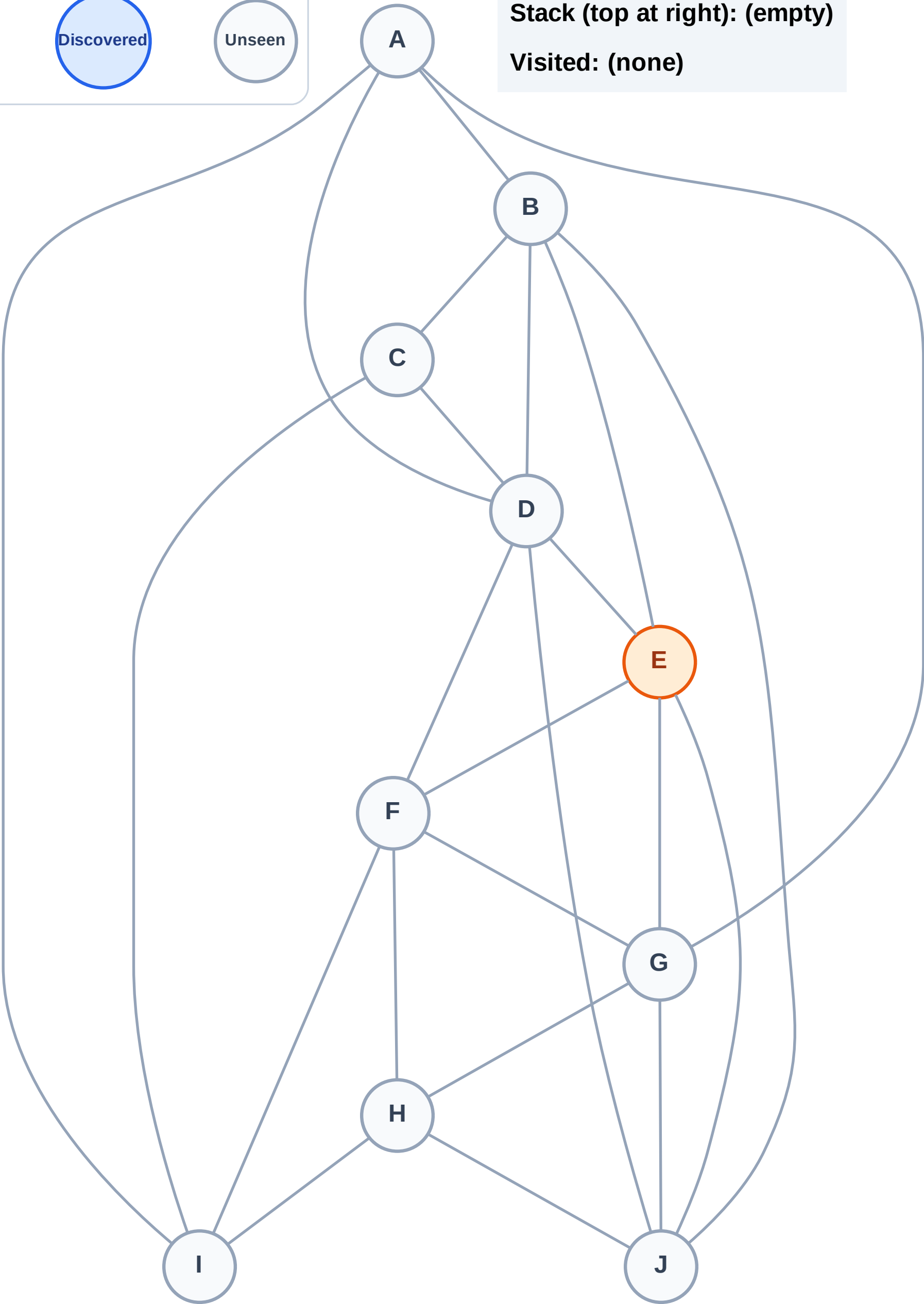
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover J, G, F, D, B from E

Legend

Complete

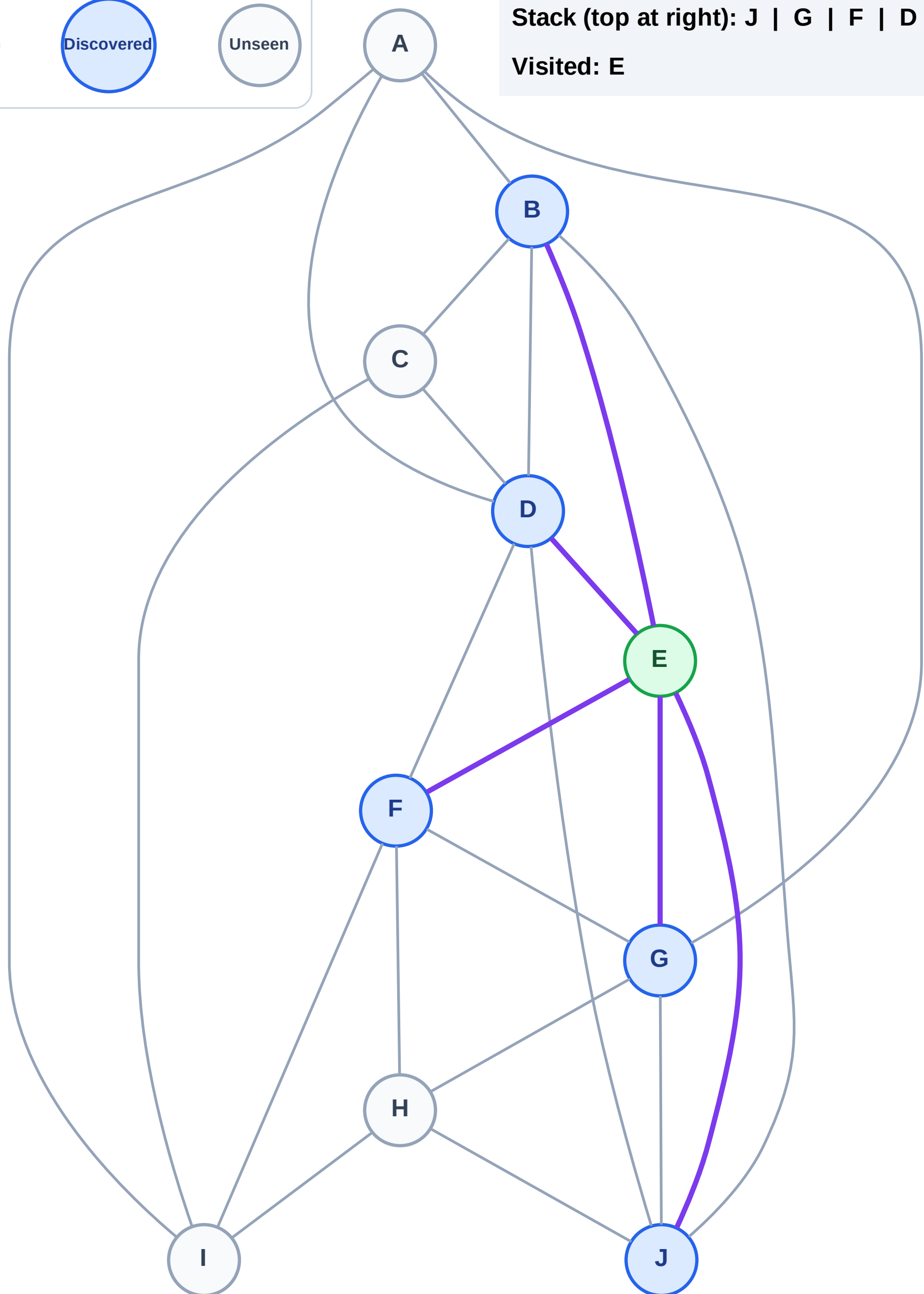
Processing

Discovered

Unseen

Stack (top at right): J | G | F | D | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

Complete

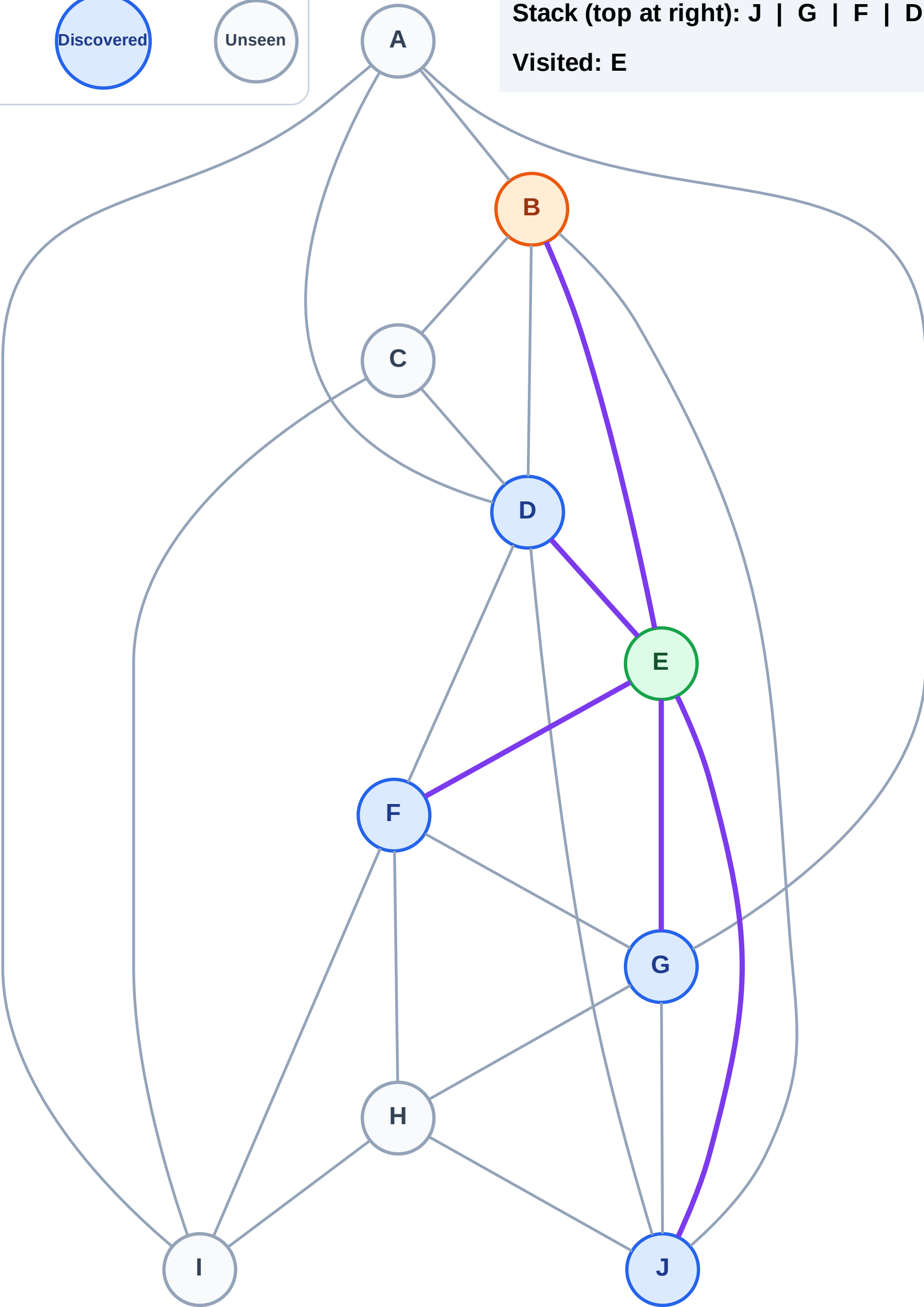
Processing

Discovered

Unseen

Stack (top at right): J | G | F | D

Visited: E



DFS Traversal

Step 5: Discover C, A from B

Legend

Complete

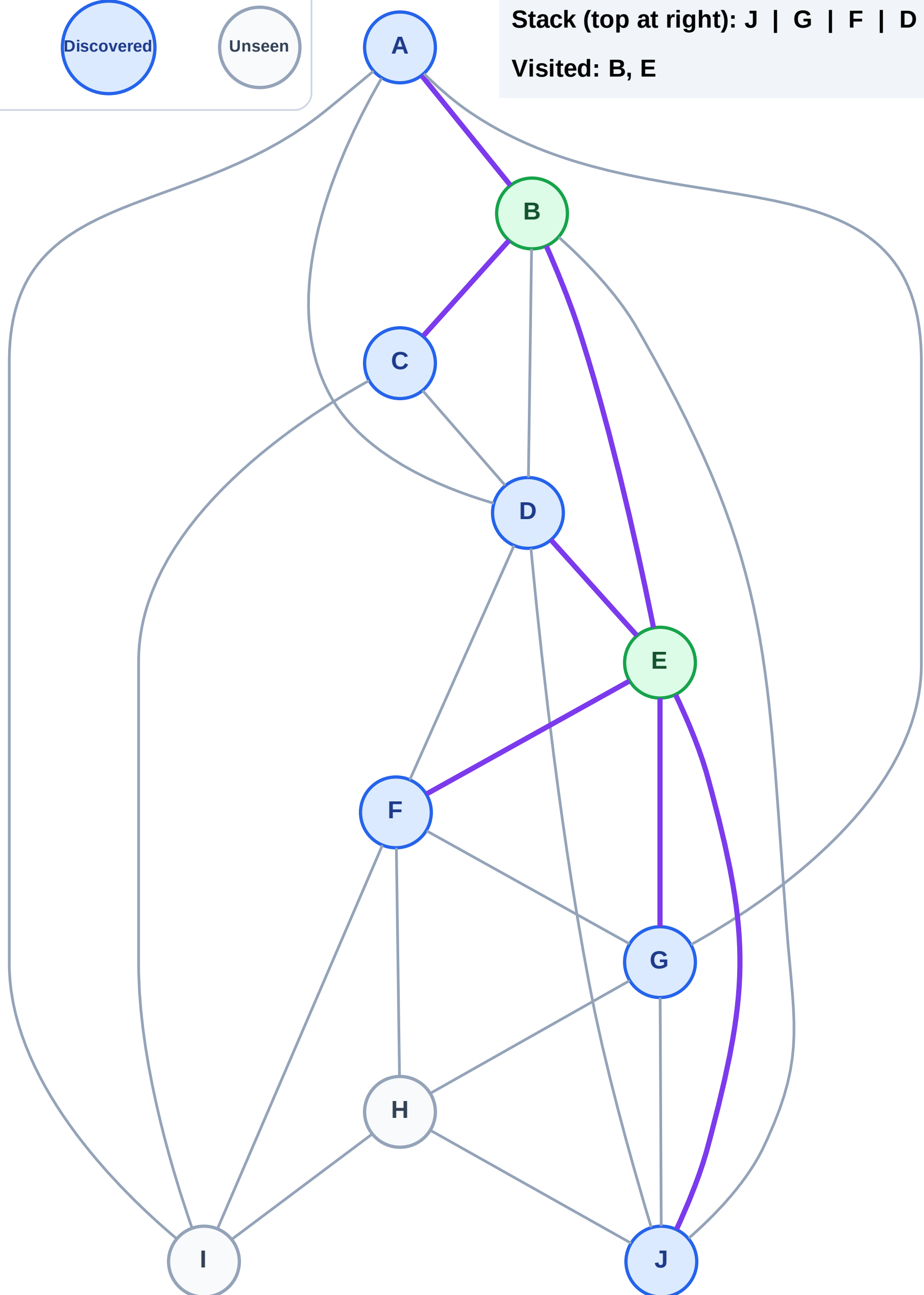
Processing

Discovered

Unseen

Stack (top at right): J | G | F | D | C | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

Complete

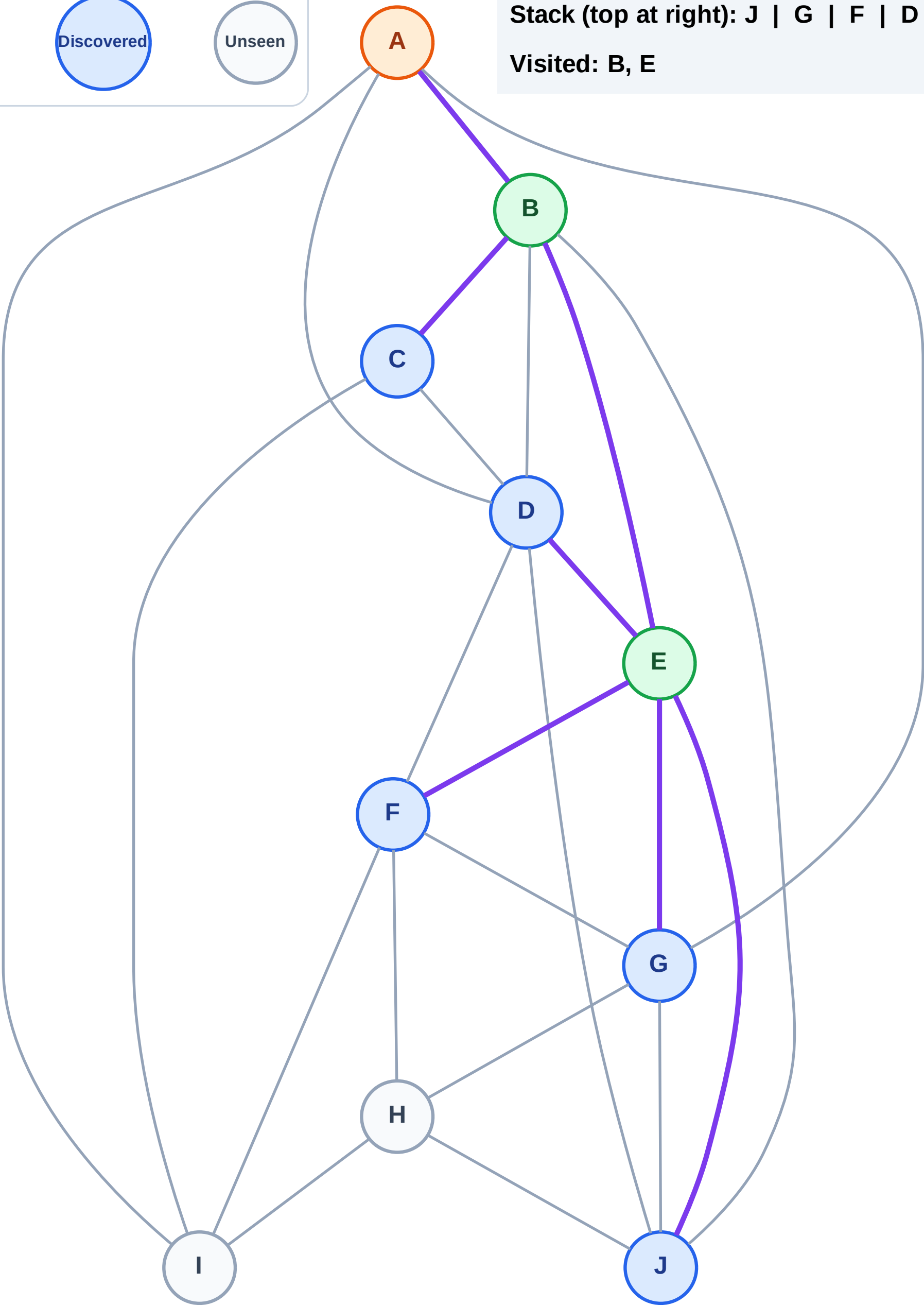
Processing

Discovered

Unseen

Stack (top at right): J | G | F | D | C

Visited: B, E



DFS Traversal

Step 7: Discover I from A

Legend

Complete

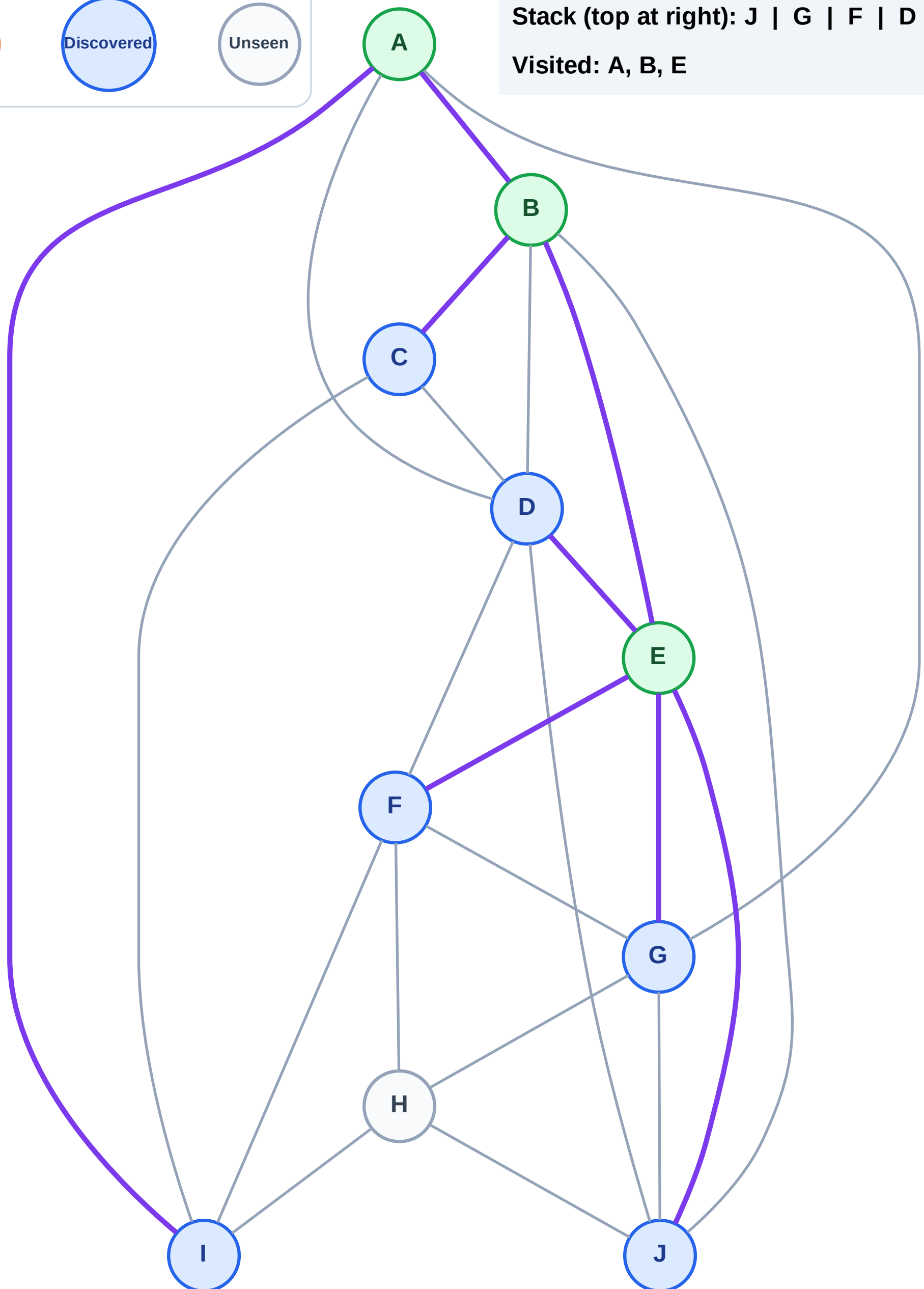
Processing

Discovered

Unseen

Stack (top at right): J | G | F | D | C | I

Visited: A, B, E



DFS Traversal

Step 8: Process node I

Legend

Complete

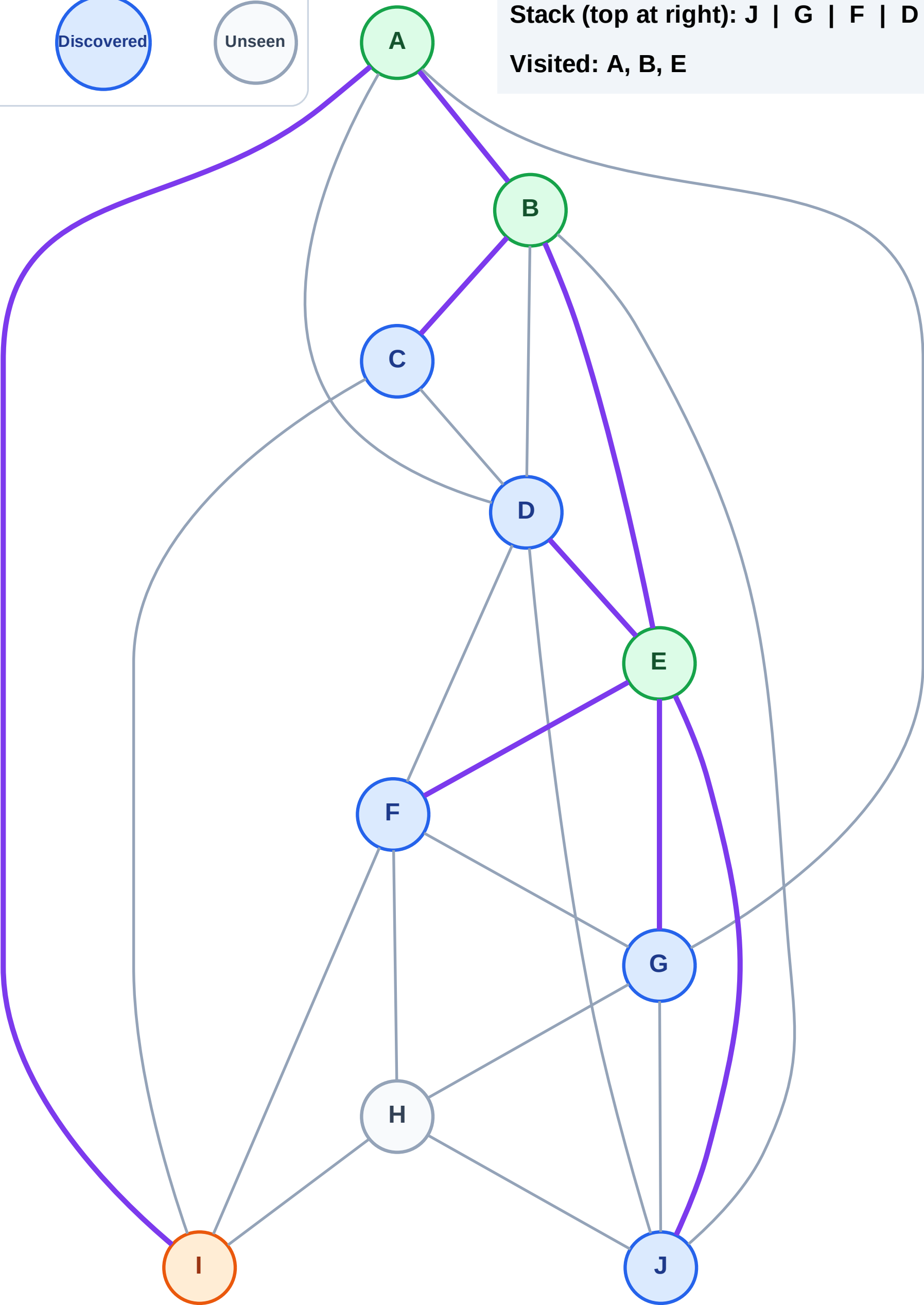
Processing

Discovered

Unseen

Stack (top at right): J | G | F | D | C

Visited: A, B, E



DFS Traversal

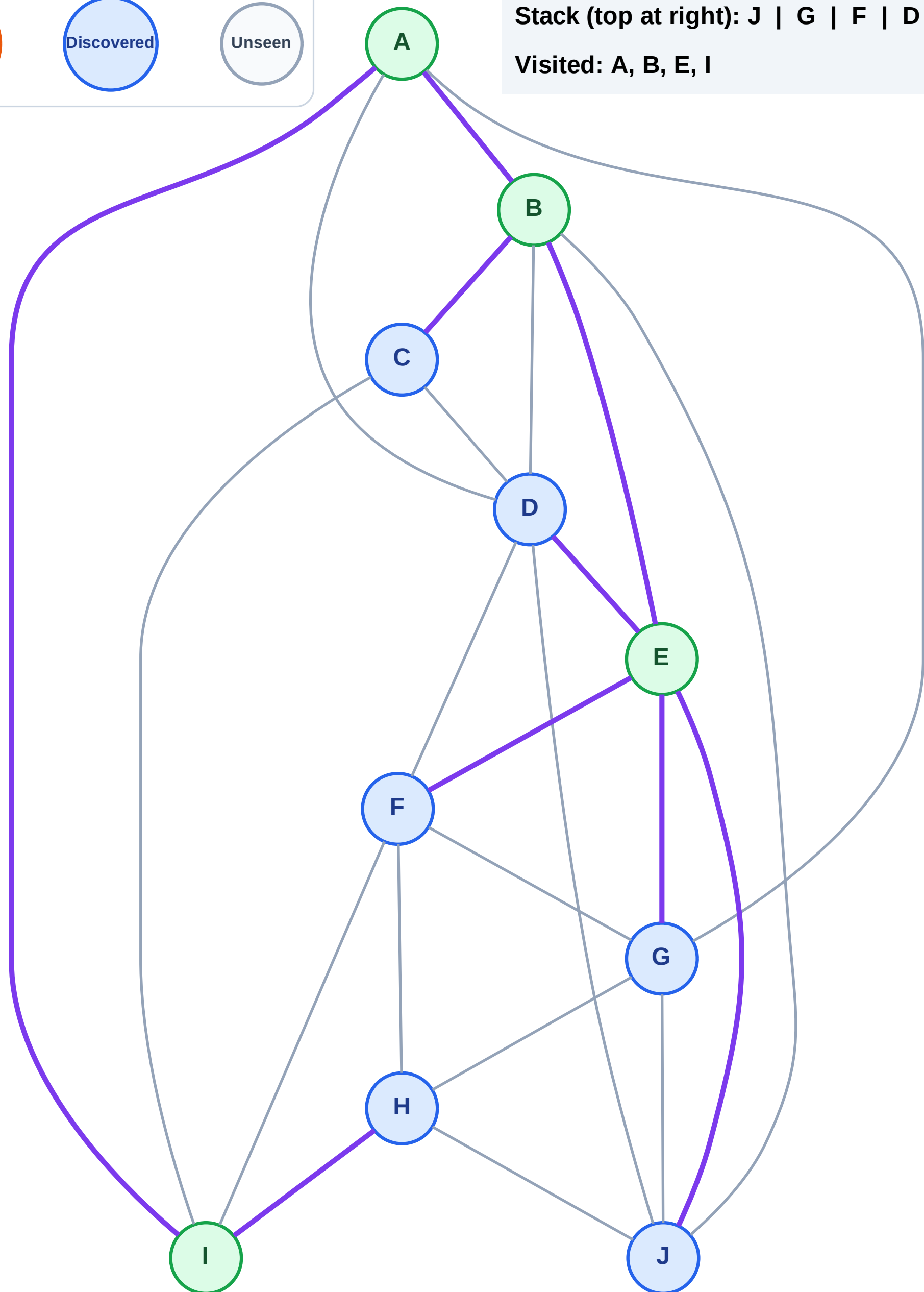
Step 9: Discover H from I

Legend



Stack (top at right): J | G | F | D | C | H

Visited: A, B, E, I



DFS Traversal

Step 10: Process node H

Legend

Complete

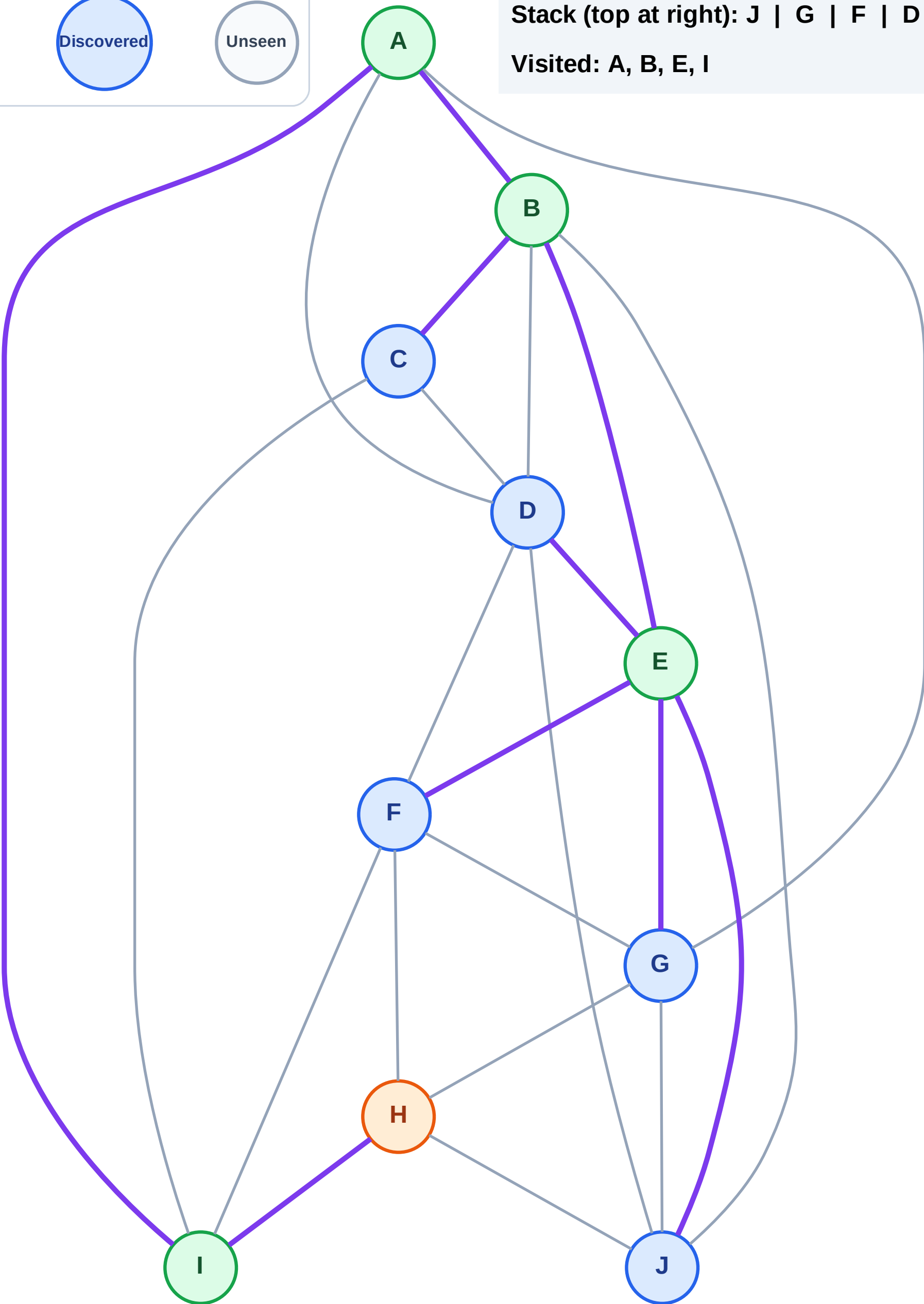
Processing

Discovered

Unseen

Stack (top at right): J | G | F | D | C

Visited: A, B, E, I



DFS Traversal

Step 11: No new neighbors from H

Legend

Complete

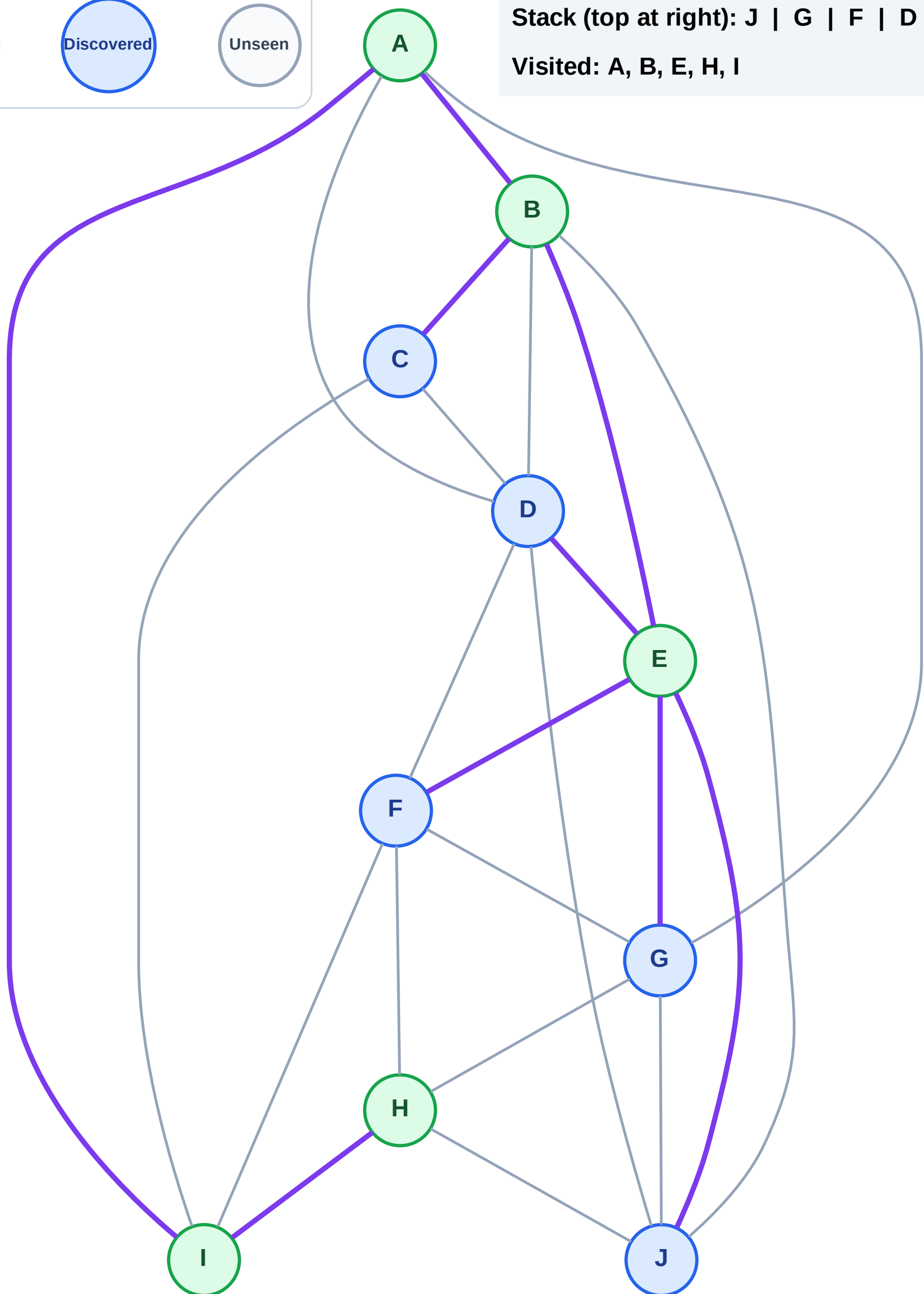
Processing

Discovered

Unseen

Stack (top at right): J | G | F | D | C

Visited: A, B, E, H, I



DFS Traversal

Step 12: Process node C

Legend

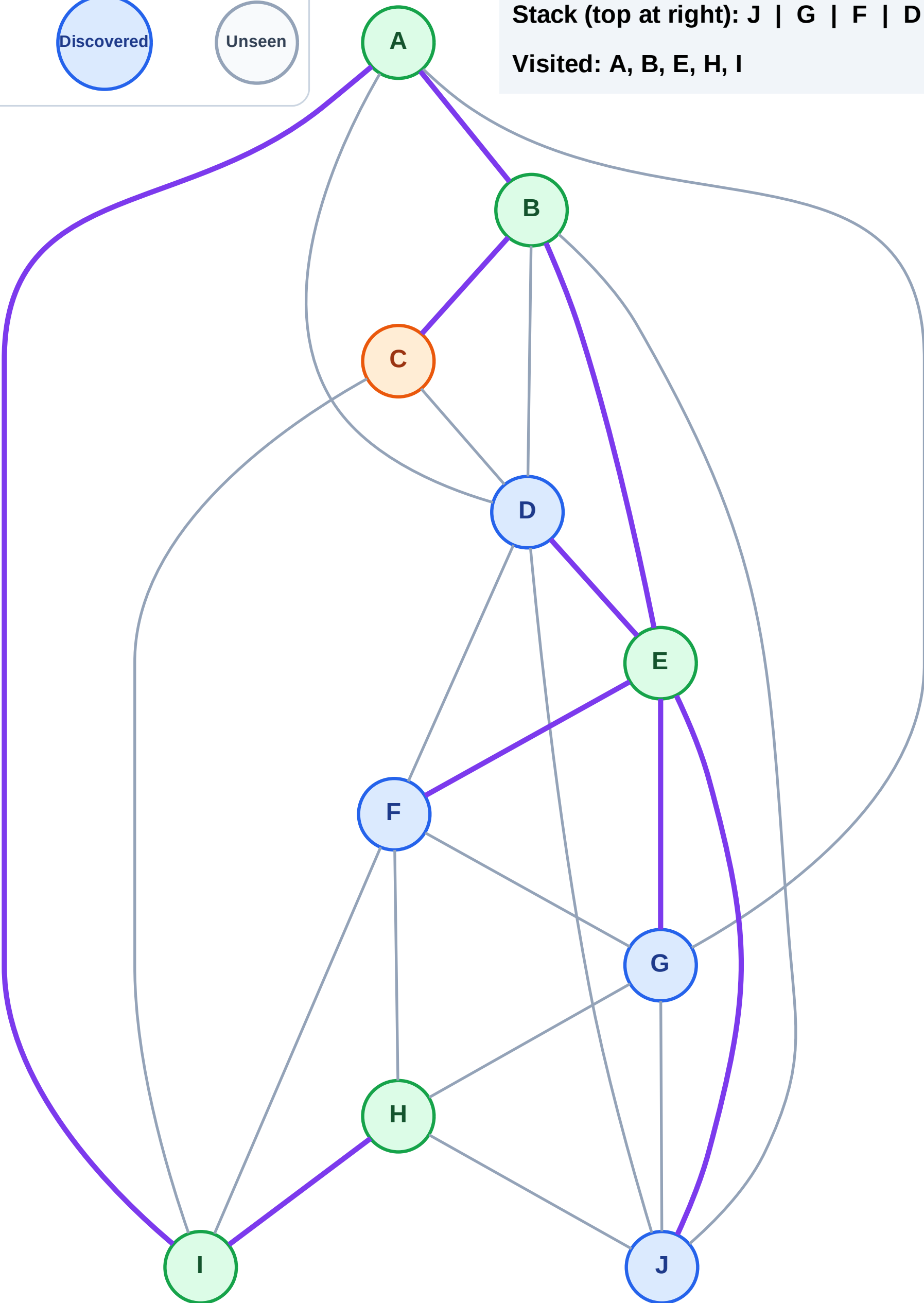
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G | F | D
Visited: A, B, E, H, I



DFS Traversal

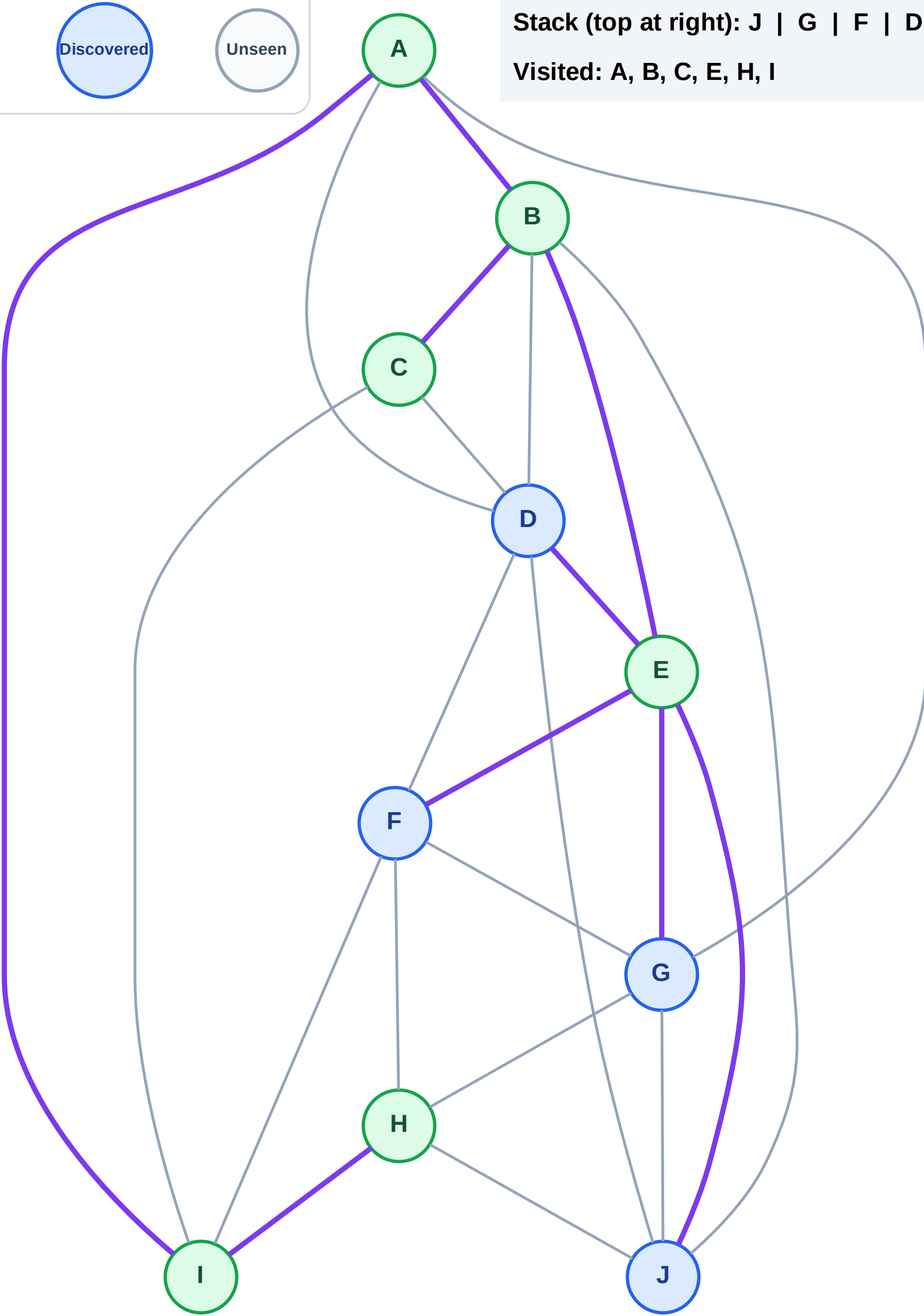
Step 13: No new neighbors from C

Legend



Stack (top at right): J | G | F | D

Visited: A, B, C, E, H, I



DFS Traversal

Step 14: Process node D

Legend

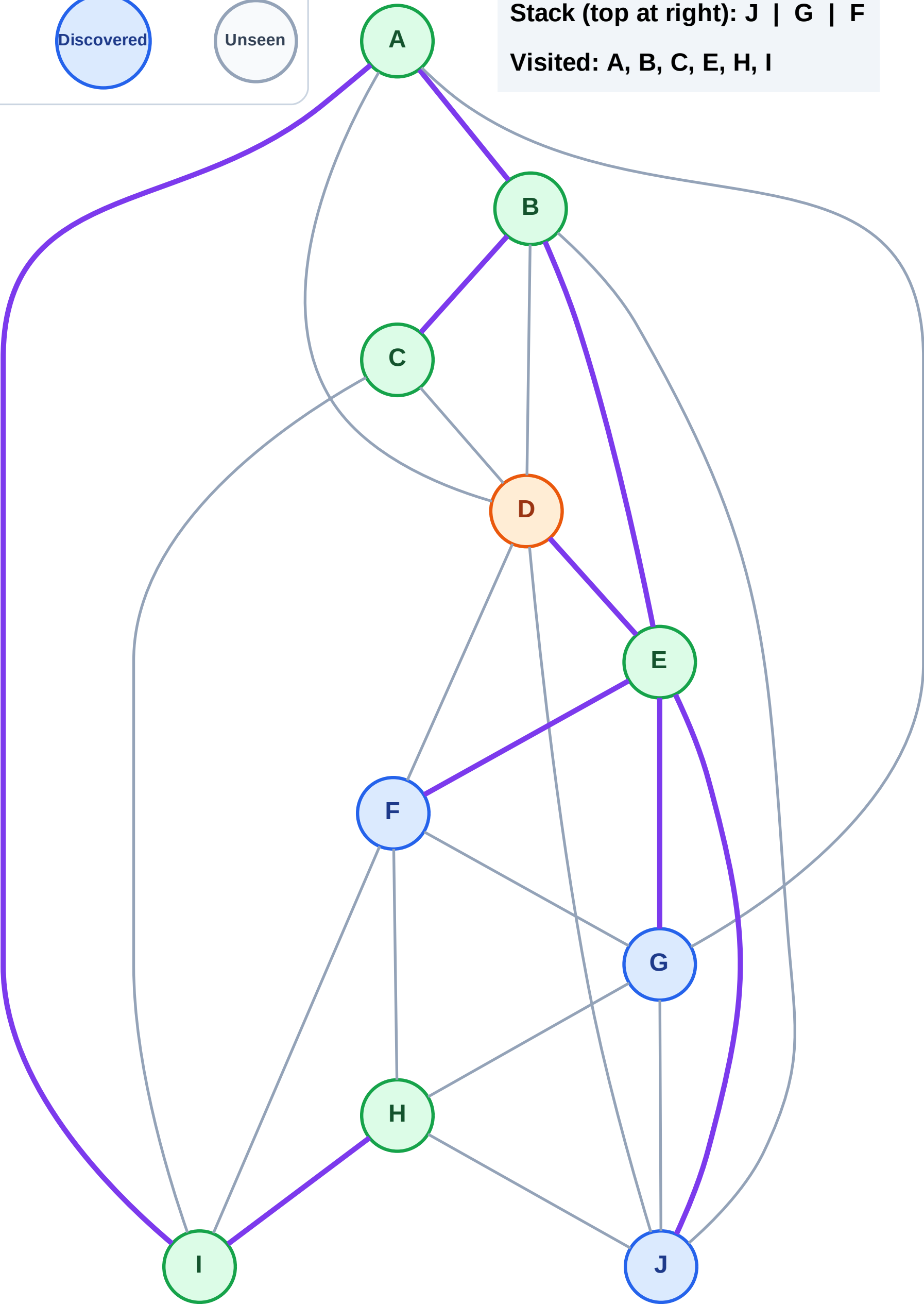
Complete

Processing

Discovered

Unseen

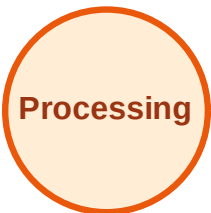
Stack (top at right): J | G | F
Visited: A, B, C, E, H, I



DFS Traversal

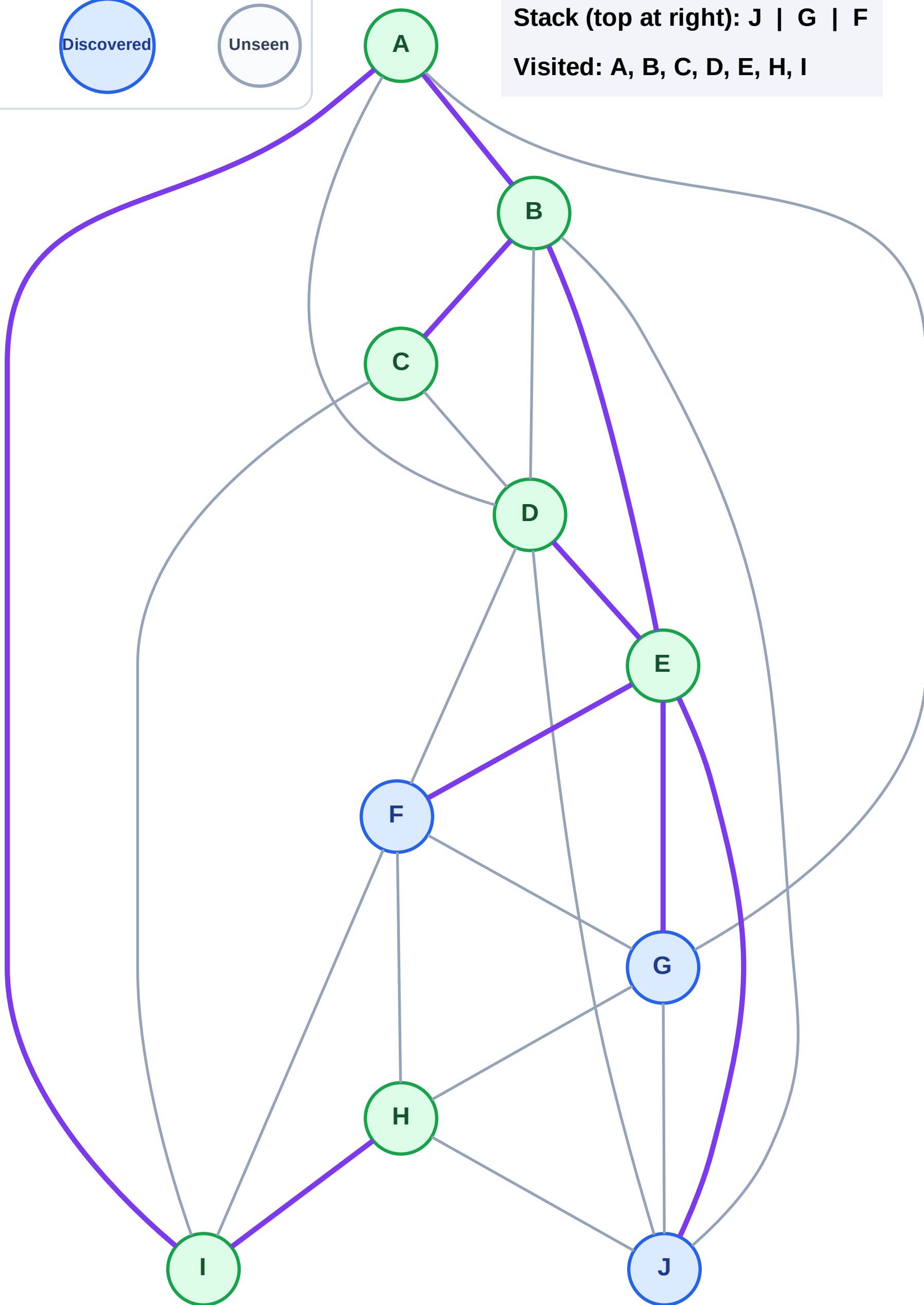
Step 15: No new neighbors from D

Legend



Stack (top at right): J | G | F

Visited: A, B, C, D, E, H, I



DFS Traversal

Step 16: Process node F

Legend

Complete

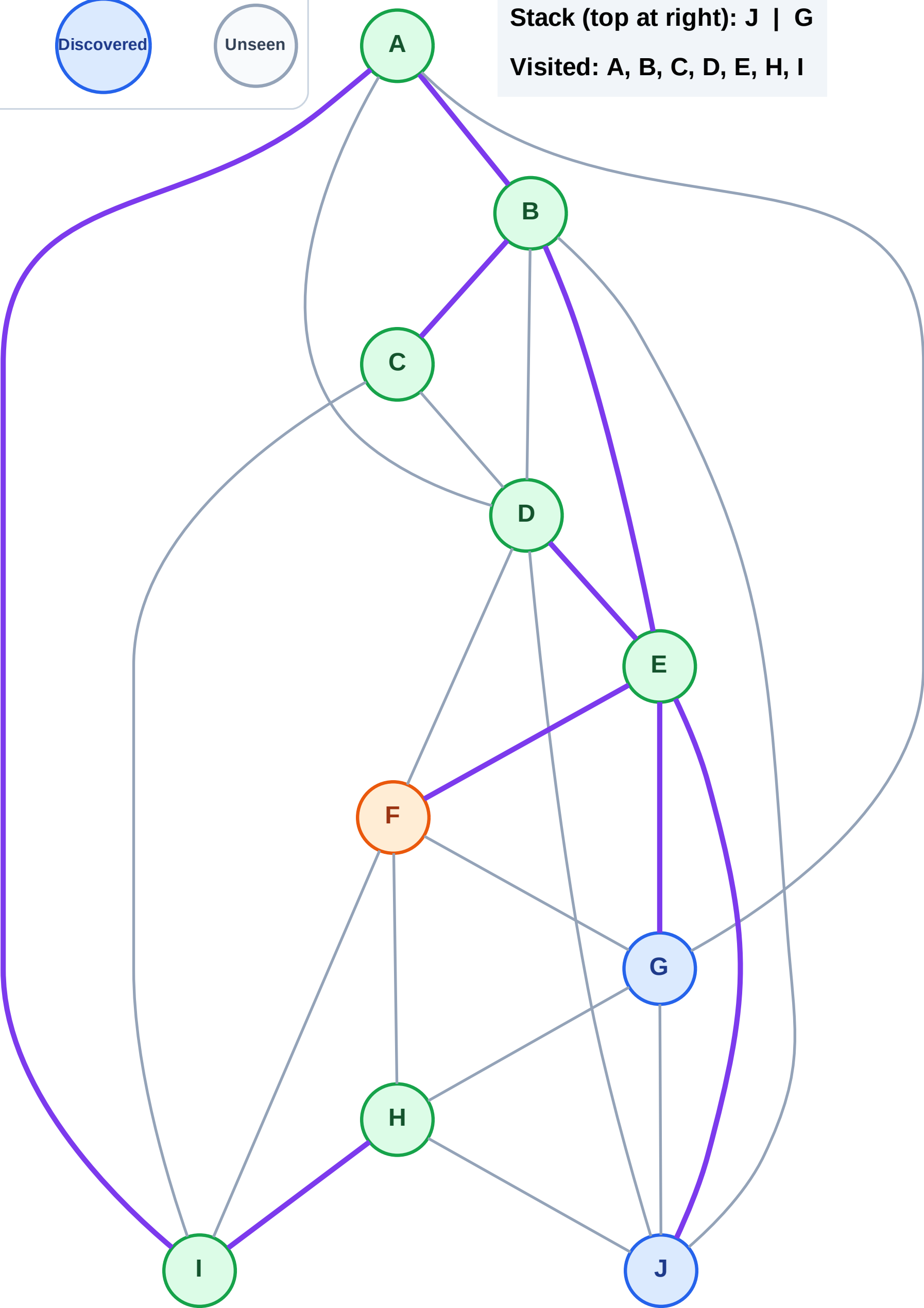
Processing

Discovered

Unseen

Stack (top at right): J | G

Visited: A, B, C, D, E, H, I



DFS Traversal

Step 17: No new neighbors from F

Legend

Complete

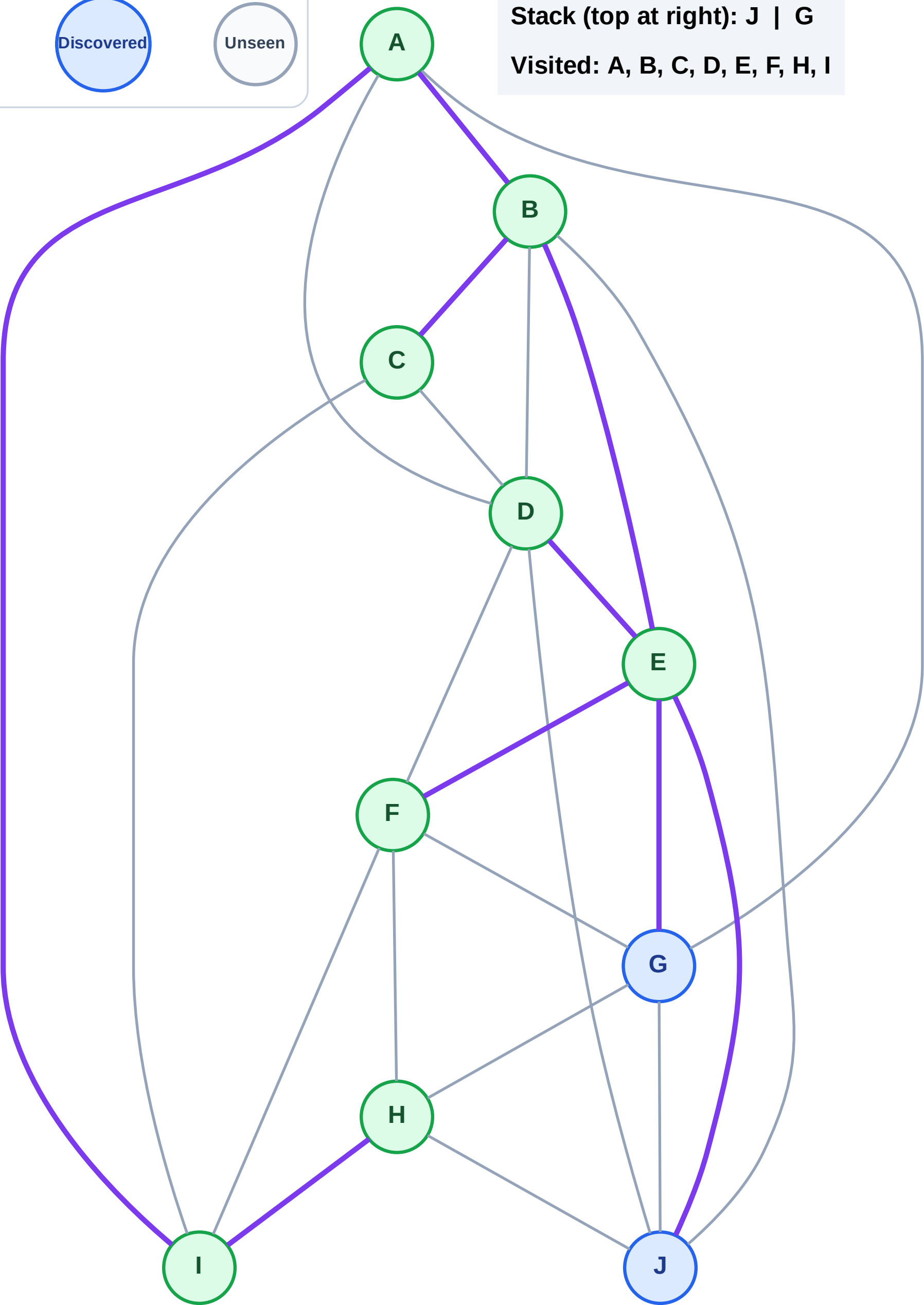
Processing

Discovered

Unseen

Stack (top at right): J | G

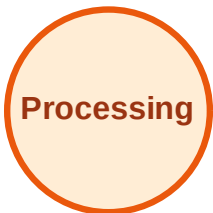
Visited: A, B, C, D, E, F, H, I



DFS Traversal

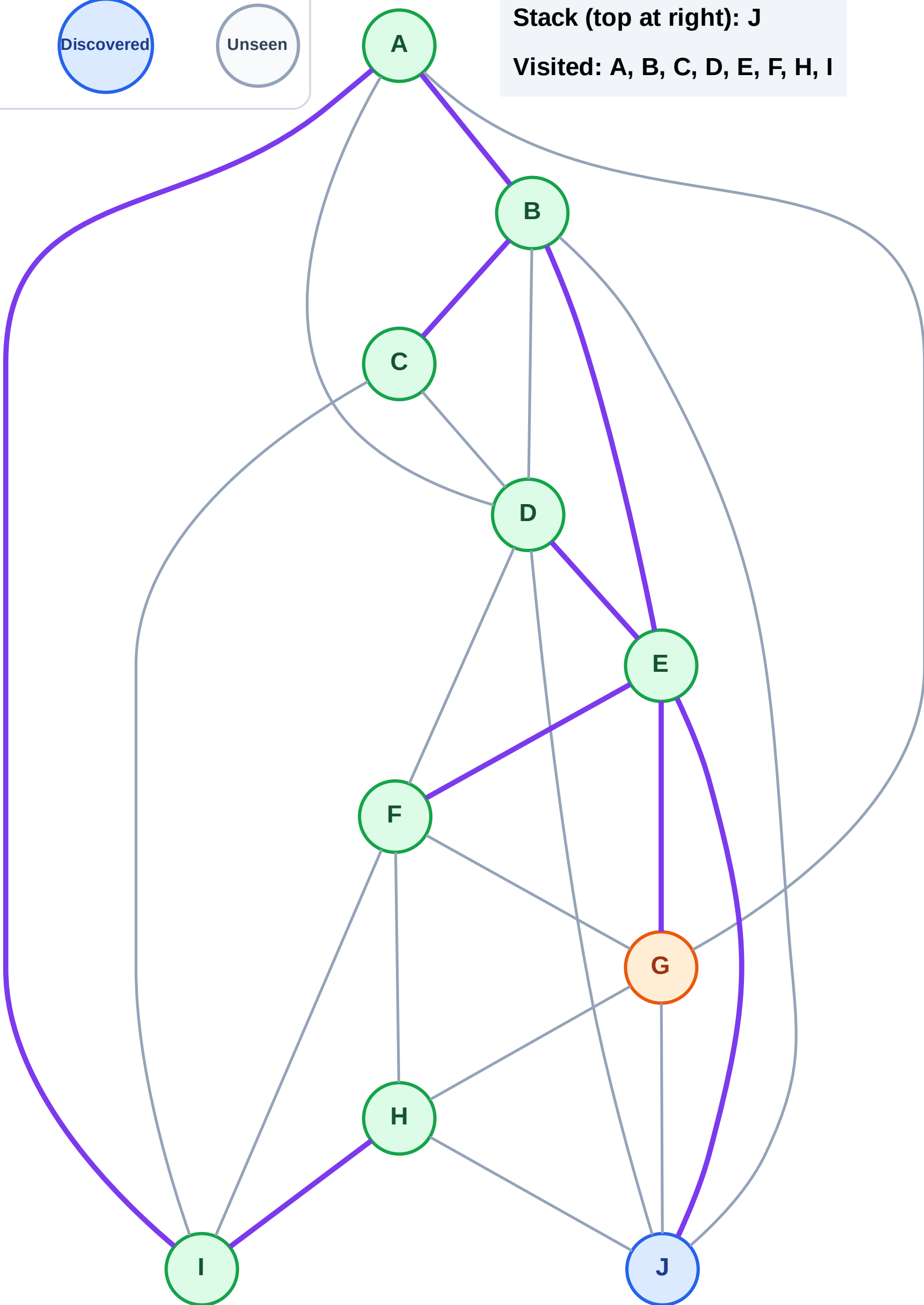
Step 18: Process node G

Legend



Stack (top at right): J

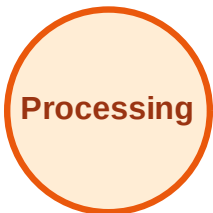
Visited: A, B, C, D, E, F, H, I



DFS Traversal

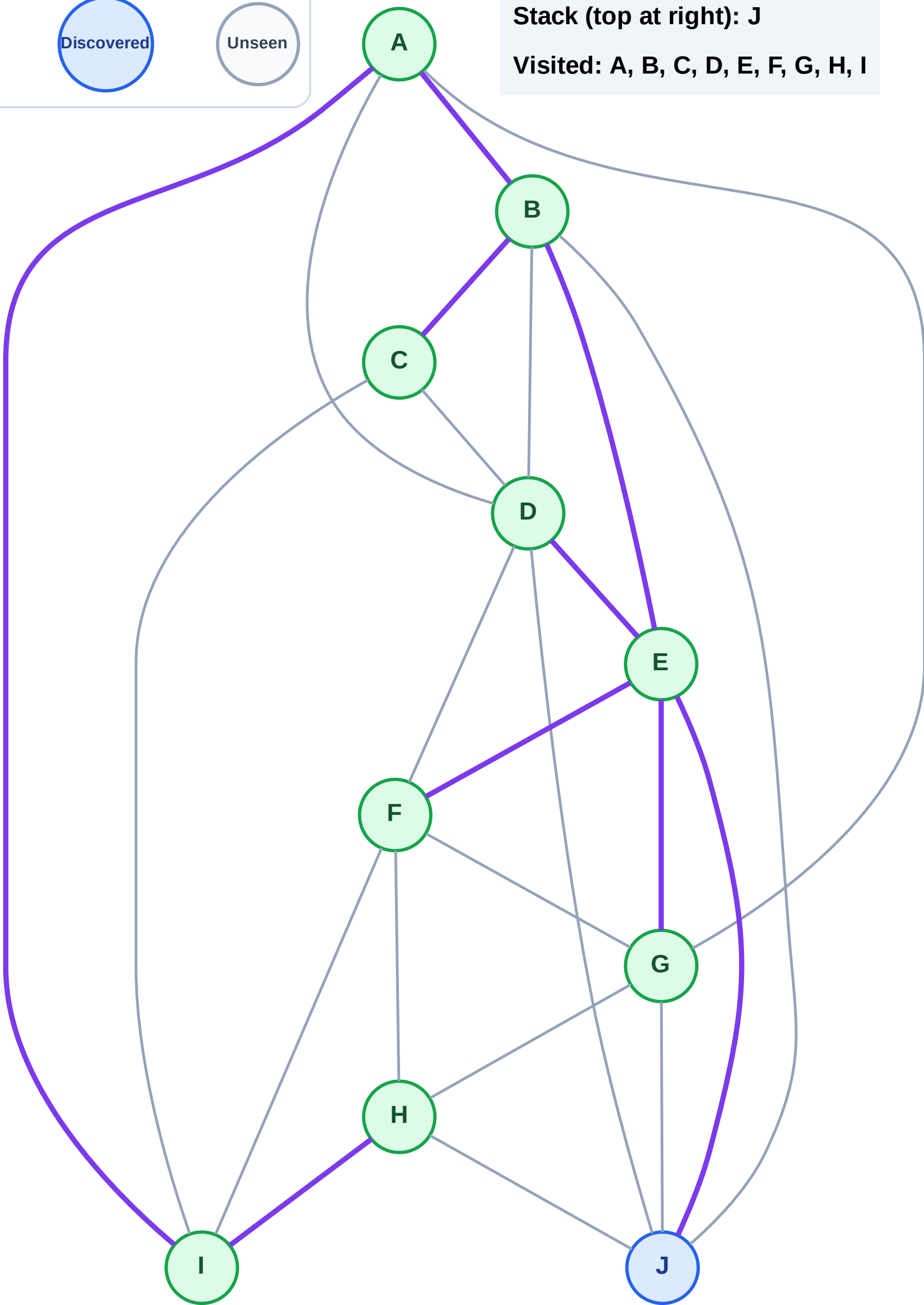
Step 19: No new neighbors from G

Legend



Stack (top at right): J

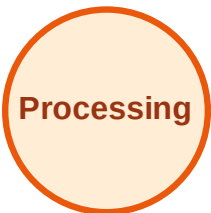
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

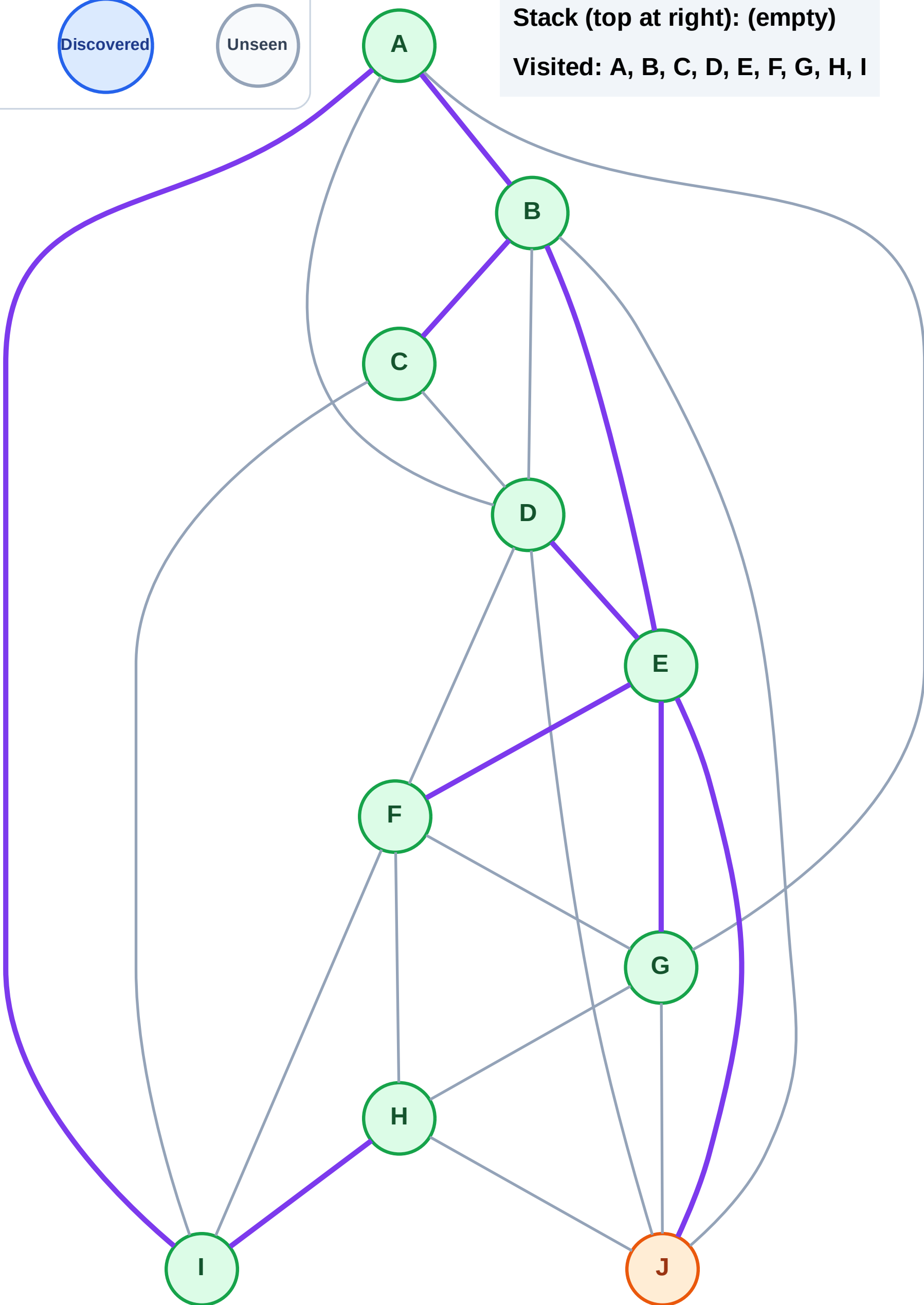
Step 20: Process node J

Legend



Stack (top at right): (empty)

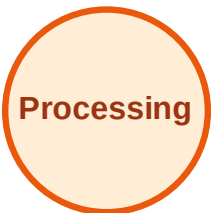
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

